



**VIDYA VIKAS INSTITUTE OF ENGINEERING AND
TECHNOLOGY, MYSORE**

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VTU Previous Question Papers

**“Information Science and Engineering”
III, V & VII Semester
Dec.2024/Jan.2025**

VVIET

Bannur road, Alanahalli, Mysore-570028

INDEX

S/L	Subject Code	Subject Title	Exam Date
1	BAD515B	Data Warehousing	Dec.2024/Jan.2025
2	BAI515A	Computer Vision	Dec.2024/Jan.2025
3	BCB501	Fundamentals of Management	Dec.2024/Jan.2025
4	BCB515A	Marketing Research and Marketing Management	Dec.2024/Jan.2025
5	BCS/BAD/BAI/BDS301	Mathematics – III for Computer Science Stream	Dec.2024/Jan.2025
6	BCS302	Digital Design and Computer Organization	Dec.2024/Jan.2025
7	BCS303	Operating Systems	Dec.2024/Jan.2025
8	BCS304	Data Structures and Applications	Dec.2024/Jan.2025
9	BCS306A	Object Oriented Programming with JAVA	Dec.2024/Jan.2025
10	BCS501	Software Engineering and Project Management	Dec.2024/Jan.2025
11	BCS502	Computer Networks	Dec.2024/Jan.2025
12	BCS503	Theory of Computation	Dec.2024/Jan.2025
13	BCS508	Environmental Studies and E – Waste Management	Dec.2024/Jan.2025
14	BCS515B	Artificial Intelligence	Dec.2024/Jan.2025
15	BESK508	Environmental Studies	Dec.2024/Jan.2025
16	BMATEC301/BEC301/BBM301	AV Mathematics III for EC/ BM Engineering	Dec.2024/Jan.2025
17	BRMK557	Research Methodology and IPR	Dec.2024/Jan.2025
18	21CS51	Automata Theory and Compiler Design	Dec.2024/Jan.2025
19	21CS52	Computer Networks	Dec.2024/Jan.2025
20	21CS53	Database Management Systems	Dec.2024/Jan.2025

21	21CS71	Big Data Analytics	Dec.2024/Jan.2025
22	21CS72	Cloud Computing	Dec.2024/Jan.2025
23	21CS735	Internet of Things	Dec.2024/Jan.2025
24	21CS741	Software Architecture and Design Patterns	Dec.2024/Jan.2025
25	21CS743	Deep Learning	Dec.2024/Jan.2025
26	21CS752	Introduction to AI and ML	Dec.2024/Jan.2025
27	21IS71	Cryptography and Network Security	Dec.2024/Jan.2025
28	21MAT755	Advanced Mathematical Methods	Dec.2024/Jan.2025
29	21RMI56	Research Methodology and Intellectual Property Rights	Dec.2024/Jan.2025
30	18CS32	Data Structures and Applications	Dec.2024/Jan.2025
31	18CS33	Analog and Digital Electronics	Dec.2024/Jan.2025
32	18CS34	Computer Organization	Dec.2024/Jan.2025
33	18CS35	Software Engineering	Dec.2024/Jan.2025
34	18CS52	Computer Networks and Security	Dec.2024/Jan.2025
35	18CS53	Database Management Systems	Dec.2024/Jan.2025
36	18CS54	Automata Theory and Computability	Dec.2024/Jan.2025
37	18CS55	Application Development using Python	Dec.2024/Jan.2025
38	18CS734	User Interface Design	Dec.2024/Jan.2025
39	18CS745	Robotic Process Automation Design and Development	Dec.2024/Jan.2025
40	18MAT31	Transform Calculus, Fourier Series and Numerical Techniques	Dec.2024/Jan.2025
41	18MAT753	Advanced Mathematical Methods	Dec.2024/Jan.2025

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Data Warehousing

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Discuss the escalating need for strategic information in modern business. How does this need impact decision making processes?	10	L2	CO2
	b.	Bring out the differences between operational systems and decision support systems in detail.	10	L3	CO2
OR					
Q.2	a.	How do you justify “Data warehousing the only viable solution?”	10	L2	CO1
	b.	Compare data warehouses and data marts. Discuss when and why an organization might choose one over the other.	10	L3	CO1
Module – 2					
Q.3	a.	Discuss on any four key issues to be considered while planning for data warehouse.	10	L2	CO2
	b.	Illustrate the process of data transformation and its significance in data warehousing	10	L2	CO2
OR					
Q.4	a.	How are data warehouse projects different from OLTP system projects? Describe any four such differences.	10	L2	CO2
	b.	With a neat diagram, explain the architectural components driven by the requirements.	10	L3	CO2
Module – 3					
Q.5	a.	Define the architectural framework and how do they interact to support data warehousing explain with a neat diagram.	10	L3	CO3
	b.	Analyze the significance of metadata in data warehousing. How does it improve data accessibly and management within the warehouse.	10	L4	CO3
OR					
Q.6	a.	List out the guidelines for selecting appropriate hardware and operating system for the data warehouse.	10	L3	CO3
	b.	With a neat diagram, discuss on the tools required for a data warehouse.	10	L4	CO3
Module – 4					
Q.7	a.	What are star schema keys, and what role do they play in a star schema. Describe how they contribute to data organization and retrieval.	10	L3	CO4
	b.	Describe the ETL tools used in data warehousing. How do these tools improve ETL process?	10	L4	CO4
OR					
Q.8	a.	Discuss the methods for handling updates to dimension tables in a data warehouse. What strategies exist for managing slowly changing dimensions?	10	L3	CO4
	b.	Outline the essential steps and requirements in the ETL process.	10	L4	CO4

Module – 5

Q.9	a.	Identify the key challenges of maintaining data quality in a data warehouse.	10	L3	CO5
	b.	List and explain the major features and functions of OLAP in data warehouse.	10	L3	CO5
OR					
Q.10	a.	List and explain five criteria for selecting information delivery tools for your data warehouse.	10	L3	CO5
	b.	Describe briefly two major features of the web –enabled data warehouse.	10	L3	CO5

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Computer Vision

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is Compute Vision? Why is vision so difficult? Discuss the real – world examples of computer vision.	10	L1	CO1
	b.	Explain the behavior of pinhole camera under different effects using a clear illustration to show the real–world example.	10	L2	CO1
OR					
Q.2	a.	Explain the phong shading model.	8	L2	CO1
	b.	Explain the Di-chromatic reflectance model.	5	L2	CO1
	c.	What is meant by image filtering? Clearly discuss types of filter.	7	L1	CO1
Module – 2					
Q.3	a.	Differentiate between a linear spatial filter and a non-linear spatial filter. Explain why bilateral filtering is quite show compared to regular separable filtering.	10	L2	CO2
	b.	Explain the binary image processing. Obtain the distance transform $D(i, j)$ of a binary image $B(i, j)$.	10	L2	CO2
OR					
Q.4	a.	Explain the derivation of Discrete Fourier Transform (DFT) form the continuous transform of the sampled function.	10	L2	CO2
	b.	What are the geometric transformation? Explain the forward warping algorithm for transforming an image.	10	L1	CO3
Module – 3					
Q.5	a.	Give the probability density functions for Gaussian noise model and Rayleigh noise models.	10	L1	CO3
	b.	Discuss the noise reduction capabilities of the following spatial filters : i) Arithmetic mean filter ii) Geometric mean filter.	10	L1	CO3
OR					
Q.6	a.	Explain the image gradient and its properties.	10	L2	CO3
	b.	Explain the following gradient operators : i) Roberts cross –gradient operators ii) Sobel operator iii) Prewitt operator iv) Laplacian operator.	10	L2	CO3

Module – 4

Q.7	a.	Define the pseudocolor processing of digital images. Explain the graphical interpretation of the intensity slicing technique.	10	L2	CO4
	b.	Discuss the procedure for conversion from RGB color model to HIS color model.	10	L2	CO4

OR

Q.8	a.	Illustrate how full-color images are handled for a variety of image processing tasks.	10	L2	CO4
	b.	Explain the color image smoothing and sharpening procedure.	10	L2	CO4

Module – 5

Q.9	a.	Explain how morphological operations are performed between images and structuring elements.	10	L2	CO5
	b.	Write short notes on the following : i) Erosin ii) Dialation.	10	L2	CO5

OR

Q.10	a.	Explain the procedures in the “boundary following” algorithm that traces the boundary in binary image.	10	L2	CO5
	b.	What is Pattern Classification? Explain the minimum distance classifier.	10	L2	CO5

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BCB501

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Fundamentals of Management

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Discuss the functions of Management.	10	L2	CO1
	b.	Explain the Fayol's administrative management.	10	L2	CO1
OR					
Q.2	a.	Explain the nature of scope of management.	10	L2	CO1
	b.	What are the Levels of Management? Briefly explain.	10	L2	CO1
Module – 2					
Q.3	a.	What is Planning? Explain the process of planning.	10	L2	CO2
	b.	Discuss on detail the types of planning.	10	L2	CO2
OR					
Q.4	a.	Explain about MBO (Management By Objectives).	10	L2	CO2
	b.	What are the Approaches of Decision making? Explain briefly.	10	L3	CO2
Module – 3					
Q.5	a.	Explain the principles of organizing.	10	L3	CO3
	b.	Explain about i) Centralization ii) Decentralization.	10	L3	CO3
OR					
Q.6	a.	Explain about organizational structure and Design it.	10	L3	CO3
	b.	“Modern Organizations Prefer Wide Span Of Management.	10	L4	CO3
Module – 4					
Q.7	a.	Explain about the theories of Leadership.	10	L3	CO4
	b.	What is Motivation? Explain the Theories of Motivation.	10	L3	CO4
OR					
Q.8	a.	Explain the steps involved in controlling process.	10	L3	CO4
	b.	“A good control system helps an organization in many ways”. Justify the statement.	10	L4	CO4

Module – 5					
Q.9	a.	Discuss about Big Five Models of personality.	10	L3	CO5
	b.	Explain the factors that influencing perception.	10	L3	CO5
OR					
Q.10	a.	What is Attitude? Explain the components of Attitude.	10	L3	CO5
	b.	How Group Decision Making is done? Explain.	10	L3	CO5

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Marketing Research and Marketing Management

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Differentiate between marketing and selling.	7	L1	CO1
	b.	Explain the concept of customer lifetime value. Why building long term relationship with customer more profitable than focusing on one time transaction.	6	L2	CO1
	c.	Elaborate on understanding of the market place and customer needs with examples.	7	L3	CO1
OR					
Q.2	a.	Illustrate how major trends in the changing marketing landscape reshaping the way business interact with customer.	10	L2	CO1
	b.	Explain the marketing process model for creating and capturing customer value.	10	L3	CO1
Module – 2					
Q.3	a.	Apply the marketing mix strategy for software industry. Provide justification for each element.	8	L3	CO2
	b.	Identify and describe the company's micro environmental factors.	6	L3	CO2
	c.	Explain the stages in consumer behavior model.	6	L3	CO2
OR					
Q.4	a.	How does Marketing contribute to the strategic planning process of a company? Explain.	8	L2	CO2
	b.	Elaborate characteristics of Business markets.	6	L2	CO2
	c.	Compare and contrast between institutional and government markets.	6	L2	CO2
Module – 3					
Q.5	a.	What is Targeting? Explain the different levels of market targeting in detail.	5	L2	CO3
	b.	What is new product development strategy? Explain the steps in the new product development process with example.	8	L2	CO3
	c.	What is brand equity? Explain branding strategies with example.	7	L3	CO3

OR					
Q.6	a.	Elaborate on product life cycle and strategies involved in each stage.	12	L2	CO3
	b.	Illustrate how a service characteristic affects its marketing strategy in an event management company.	8	L3	CO3
Module – 4					
Q.7	a.	Explain the factors that influence marketing research decisions.	6	L2	CO4
	b.	What do you mean Research Design? Explain the different types of research.	8	L3	CO4
	c.	Explain the issues in International Research Design.	6	L3	CO4
OR					
Q.8	a.	Elaborate on Marketing Research Process with examples.	8	L3	CO4
	b.	What is Marketing Research? Explain the role of marketing in managerial decision making.	6	L3	CO4
	c.	What is Research Approach? Provide categories in research approaches.	6	L3	CO4
Module – 5					
Q.9	a.	What are the internal and external sources of secondary data? Provide examples for each.	7	L2	CO5
	b.	Ascertain benefits and limitations of secondary data.	7	L3	CO5
	c.	What are qualitative methods? Explain the characteristics of qualitative research.	6	L3	CO5
OR					
Q.10	a.	Ascertain the issues faced in data collection.	6	L5	CO5
	b.	What are the influencing the choice of statistical techniques? Discuss.	7	L4	CO5
	c.	Explain the primary data collection methods.	7	L3	CO5

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Mathematics – III for Computer Science Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Mathematics Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1									M	L	C		
Q.1	a.	A random variable x has the following prob. density function for various values of x .							07	L2	CO1		
		x	0	1	2	3	4	5				6	7
		P(x)	0	k	2k	2k	3k	k ²				2k ²	7k ² +k
Find the value of k and evaluate $P(x < 6)$, $P(3 < x \leq 6)$ and $(x \geq 6)$.													
	b.	Derive the mean and variance of Poisson distribution.							06	L2	CO2		
	c.	In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for? (i) less than 10 minutes (ii) more than 10 minutes and (iii) between 10 and 12 minutes.							07	L3	CO2		
OR													
Q.2	a.	The probability density function of $f(x) = \begin{cases} Kx^2, & -3 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of K and evaluate (i) $P(x < 2)$, $P(x > 1)$ (ii) $P(1 \leq x \leq 2)$							07	L3	CO1		
	b.	When a coin is tossed 4 times, find the probability of getting (i) exactly one head (ii) atleast three heads and (iii) less than two heads.							06	L2	CO2		
	c.	The marks of 1000 students in an examination follows a normal distribution with mean > 0 and S.D 5. Find the number of students whose marks will be (i) less than 65 (ii) more than 75 and (iii) between 65 and 75.							07	L2	CO2		
Module – 2													
Q.3	a.	If the joint probability distribution of x and y is given by $f(x, y) = \frac{1}{30}(x + y), \text{ for } x = 0, 1, 2, 3; y = 0, 1, 2$ Find (i) $P(x \leq 2, y = 1)$ (ii) $P(x > y)$							07	L2	CO2		
	b.	Find the unique fixed probability vector of $P = \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}$							06	L2	CO3		
	c.	Three boys A, B and C are throwing a ball to each other. A always throw the ball to B. B always throw the ball to A and C is just as likely to throw the ball to A as to B. Find the probability that C has the ball after three throws, if C starts the game.							07	L3	CO3		

OR

OR																						
Q.4	a.	The joint prob. distribution for the following data, find E(x) and E(y). <table><tr><td>Y \ X</td><td>-2</td><td>-1</td><td>4</td><td>5</td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.0</td><td>0.3</td></tr><tr><td>2</td><td>0.2</td><td>0.1</td><td>0.1</td><td>0</td></tr></table>			Y \ X	-2	-1	4	5	1	0.1	0.2	0.0	0.3	2	0.2	0.1	0.1	0	07	L2	CO2
Y \ X	-2	-1	4	5																		
1	0.1	0.2	0.0	0.3																		
2	0.2	0.1	0.1	0																		
	b.	Show that the matrix $P = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$ is a regular stochastic matrix.			06	L2	CO3															
	c.	A gambler's luck follows pattern. If he wins a game the prob. of winning the next game is 0.6. However, if he loses a game, the prob. of losing the next game is 0.7. There is an even chance of the gambler winning the first game. What is the prob. of he winning the second game.			07	L3	CO3															
Module – 3																						
Q.5	a.	Define (i) Null hypothesis (ii) A statistic (iii) Standard error (iv) Level of significance (v) Test of significance.			07	L1	CO4															
	b.	A coin was tossed 400 times and head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% LOS.			06	L3	CO4															
	c.	In a city A 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant at 5% significance level?			07	L3	CO5															
OR																						
Q.6	a.	Explain the following terms: (i) Type-I and Type-II errors (ii) Statistical hypothesis (iii) Critical region (iv) Alternate hypothesis			07	L1	CO4															
	b.	The average marks in Engg. Maths of a sample of 100 students was 51 with S.D 6 marks. Could this have been a random sample from a population with average marks 50?			06	L2	CO5															
	c.	One type of aircraft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of aircrafts so far as engine defects are concerned? Test at 0.05 significance level.			07	L3	CO4															
Module – 4																						
Q.7	a.	State central limit theorem. Use the theorem to evaluate $P(50 < x < 56)$ where $\bar{x}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$.			07	L2	CO4															
	b.	Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find 95% confidence interval for the population mean. Given that $Z(0.15) = 0.0596$.			06	L2	CO5															
	c.	Fit a Poisson distribution to the following data and test for goodness of fit at 5% LOS. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f</td><td>419</td><td>352</td><td>154</td><td>56</td><td>19</td></tr></table>			x	0	1	2	3	4	f	419	352	154	56	19	07	L3	CO5			
x	0	1	2	3	4																	
f	419	352	154	56	19																	

OR

Q.8	a.	Height of a random sample of 50 college student showed a mean of 174.5 cms and a S.D 6.9 cms. Construct 99% confidence limits for the mean height of all college students.	07	L2	CO4
	b.	A random sample of 10 boys had the following I.Q : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. DO these data support the assumption of a population mean I.Q of 100 (at 5% LOS)?	06	L3	CO5
	c.	The theory predicts the propositions of beans in the four groups, G_1 , G_2 , G_3 , G_4 should be in the ratio 9 : 3 : 3 : 1. In experiment with 1600 beans the numbers in the groups were 882, 313, 287 and 118. Does the experimental support the theory.	07	L3	CO5

Module – 5

Q.9	a.	The varieties of wheat A, B, C were shown in four plots each and the following yields in quintals per acre were obtained. <table><tr><td>A</td><td>8</td><td>4</td><td>6</td><td>7</td></tr><tr><td>B</td><td>7</td><td>6</td><td>5</td><td>3</td></tr><tr><td>C</td><td>2</td><td>5</td><td>4</td><td>4</td></tr></table> Test the significance of difference between the yields of varieties, given that 5% tabulated value of $F = 4.26$ with (2, 9) d.f. Set up one-way ANOVA and using direct method.	A	8	4	6	7	B	7	6	5	3	C	2	5	4	4	10	L3	CO6										
A	8	4	6	7																										
B	7	6	5	3																										
C	2	5	4	4																										
	b.	Present your conclusion after doing ANOVA to the following results of the Latin-square design conducted in respect of five fertilizers which were used on plots of different fertility. <table><tr><td>A(16)</td><td>B(10)</td><td>C(11)</td><td>D(9)</td><td>E(9)</td></tr><tr><td>E(10)</td><td>C(9)</td><td>A(14)</td><td>B(12)</td><td>D(11)</td></tr><tr><td>B(15)</td><td>D(8)</td><td>E(8)</td><td>C(10)</td><td>A(18)</td></tr><tr><td>D(12)</td><td>E(6)</td><td>B(13)</td><td>A(13)</td><td>C(12)</td></tr><tr><td>C(13)</td><td>A(11)</td><td>D(10)</td><td>E(7)</td><td>B(14)</td></tr></table>	A(16)	B(10)	C(11)	D(9)	E(9)	E(10)	C(9)	A(14)	B(12)	D(11)	B(15)	D(8)	E(8)	C(10)	A(18)	D(12)	E(6)	B(13)	A(13)	C(12)	C(13)	A(11)	D(10)	E(7)	B(14)	10	L3	CO6
A(16)	B(10)	C(11)	D(9)	E(9)																										
E(10)	C(9)	A(14)	B(12)	D(11)																										
B(15)	D(8)	E(8)	C(10)	A(18)																										
D(12)	E(6)	B(13)	A(13)	C(12)																										
C(13)	A(11)	D(10)	E(7)	B(14)																										

OR

Q.10	a.	Set up two-way ANOVA table for the data given below, using coding method subtracting 40 from the given numbers. <table><tr><th rowspan="2">Pieces of land</th><th colspan="4">Treatment</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>P</td><td>45</td><td>40</td><td>38</td><td>37</td></tr><tr><td>Q</td><td>43</td><td>41</td><td>45</td><td>38</td></tr><tr><td>R</td><td>39</td><td>39</td><td>41</td><td>41</td></tr></table>	Pieces of land	Treatment				A	B	C	D	P	45	40	38	37	Q	43	41	45	38	R	39	39	41	41	10	L3	CO6
Pieces of land	Treatment																												
	A	B	C	D																									
P	45	40	38	37																									
Q	43	41	45	38																									
R	39	39	41	41																									
	b.	There are three main brands of a certain power. A set of its 120 sales is examined and found to be allocated among four groups (A, B, C, D) and brands (I, II, III) as follows: <table><tr><th rowspan="2">Brands</th><th colspan="4">Groups</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>I</td><td>0</td><td>4</td><td>8</td><td>15</td></tr><tr><td>II</td><td>5</td><td>8</td><td>13</td><td>6</td></tr><tr><td>III</td><td>18</td><td>19</td><td>11</td><td>13</td></tr></table> Is there any significant difference in brands preference? Answer at 5% level, using one-way ANOVA. Take 10 as the code value to subtract it from all given values.	Brands	Groups				A	B	C	D	I	0	4	8	15	II	5	8	13	6	III	18	19	11	13	10	L3	CO6
Brands	Groups																												
	A	B	C	D																									
I	0	4	8	15																									
II	5	8	13	6																									
III	18	19	11	13																									

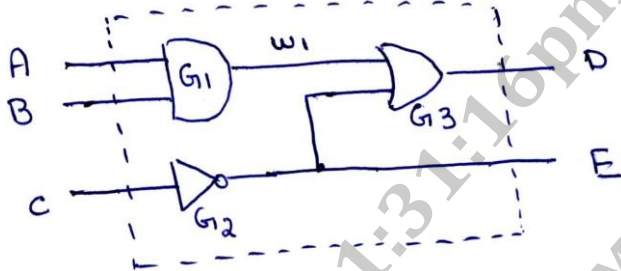
Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Digital Design and Computer Organization

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Determine the complement of the following function: (i) $F = xy' + x'y$ (ii) $F = x'yz' + x'y'z$		06	L3	CO1
	b.	Describe map method for three variables.		04	L2	CO1
	c.	Apply K map technique to simplify the following function: (i) $F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$ (ii) $F(x, y, z) = x'y + yz' + y'z'$		10	L3	CO1
OR						
Q.2	a.	Apply K map technique to simplify the function : $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ and $d(w, x, y, z) = \Sigma(0, 2, 5)$		06	L3	CO1
	b.	Determine all the prime implicants for the Boolean function F and also determine which are essential $F(w, x, y, z) = \Sigma(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$		10	L3	CO1
	c.	Develop a verilog gate-level description of the circuit shown in Fig.Q2(c). 		04	L3	CO1
Module – 2						
Q.3	a.	Explain the combinational circuit design procedure with code conversion example.		10	L2	CO2
	b.	Design a full adder circuit. Also develop data flow verilog model for full adder.		10	L3	CO2
OR						
Q.4	a.	Describe 4×1 MUX with block diagram and truth table. Also develop a behavioral model verilog code for 4×1 MUX.		10	L2	CO2
	b.	What are storage elements? Explain the working of SR and D latch along with logic diagram and function table.		10	L2	CO2
Module – 3						
Q.5	a.	Explain the basic operational concepts between the processor and memory.		10	L2	CO3
	b.	Describe the following: (i) Processor clock (ii) Basic performance equation (iii) Clock rate (iv) SPEC rating		10	L2	CO3
OR						
Q.6	a.	Define addressing mode. Explain any four types of addressing mode with example.		10	L2	CO3

	b.	Mention four types of operations to be performed by instructions in a computer. Explain the basic types of instruction formats to carry out. $C \leftarrow [A] + [B]$	10	L2	CO3
Module – 4					
Q.7	a.	With a neat diagram, explain the concept of accessing I/O devices.	10	L2	CO4
	b.	What is bus arbitration? Explain centralized and distributed arbitration method with a neat diagram.	10	L2	CO4
OR					
Q.8	a.	With neat sketches, explain various methods for handling multiple interrupts requests raised by multiple devices.	10	L2	CO4
	b.	What is cache memory? Explain any two mapping function of cache memory.	10	L2	CO4
Module – 5					
Q.9	a.	Draw the single bus architecture and write the control sequence for execution of instruction ADD (R ₃), R ₁ .	10	L3	CO5
	b.	With suitable diagram, explain the concept of register transfer and fetching of word from memory.	10	L2	CO5
OR					
Q.10	a.	With a neat diagram, explain the flow of 4-stage pipeline operation.	10	L2	CO5
	b.	Explain the role of cache memory and pipeline performance.	10	L2	CO5

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Operating Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C												
Q.1	a.	Define Operating System. Explain dual mode of operating systems with a neat diagram.	06	L1 L2	CO1												
	b.	Distinguish between the following terms: i) Multiprogramming and Multitasking ii) Multiprocessor and Clustered system	06	L2	CO1												
	c.	Explain with a neat diagram VM-WARE Architecture.	08	L1 L2	CO1												
OR																	
Q.2	a.	List and explain the services provided by OS for the user and efficient operation of system.	06	L2	CO1												
	b.	Explain the different computing equipments.	06	L2	CO1												
	c.	What are systems calls? List and explain the different types of systems calls.	08	L1 L2	CO1												
Module – 2																	
Q.3	a.	What is process? Explain process state diagram and process control block with a neat diagram.	10	L1 L2	CO2												
	b.	What is interprocess communication? Explain direct and indirect communication with respect to message passing system.	10	L1 L2	CO2												
OR																	
Q.4	a.	List and explain the different types of multithreading models.	06	L1 L2	CO2												
	b.	Calculate the average waiting time and average turnaround time by drawing the Gantt-chart using FCFS, SJF, RR (Q = 4ms) and priority scheduling (Higher Number is having highest priority). <table border="1"><tr><td>Process</td><td>B.T. (ms)</td><td>Priority</td></tr><tr><td>P₁</td><td>24</td><td>1</td></tr><tr><td>P₂</td><td>03</td><td>2</td></tr><tr><td>P₃</td><td>03</td><td>3</td></tr></table>	Process	B.T. (ms)	Priority	P ₁	24	1	P ₂	03	2	P ₃	03	3	14	L3	CO2
Process	B.T. (ms)	Priority															
P ₁	24	1															
P ₂	03	2															
P ₃	03	3															
Module – 3																	
Q.5	a.	What is critical section? Give the Peterson’s solution to 2 processes critical section problem.	05	L1 L2	CO3												
	b.	Explain Reader’s and Writer’s problem in detail.	07	L2	CO3												
	c.	What is semaphore? Discuss the solution to the classical dinning philosopher problem.	08	L1 L2	CO3												

OR																																																																													
Q.6	a.	What is a Deadlock? What are the necessary conditions for the deadlock to occur?				06	L1 L2	CO3																																																																					
	b.	Consider the following snap shot of the system. <table><tr><th rowspan="2">Process</th><th colspan="3">Allocation</th><th colspan="3">Max</th><th colspan="3">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr><tr><td>P₀</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P₁</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table> Answer the following questions: i) What is the content of the matrix need? ii) Is the system on a safe state? If so, find safe sequence. iii) If P ₁ requirements for (1, 0, 2) additional resources can P ₁ be granted.				Process	Allocation			Max			Available			A	B	C	A	B	C	A	B	C	P ₀	0	1	0	7	5	3	3	3	2	P ₁	2	0	0	3	2	2				P ₂	3	0	2	9	0	2				P ₃	2	1	1	2	2	2				P ₄	0	0	2	4	3	3				14	L3	CO2
Process	Allocation			Max			Available																																																																						
	A	B	C	A	B	C	A	B	C																																																																				
P ₀	0	1	0	7	5	3	3	3	2																																																																				
P ₁	2	0	0	3	2	2																																																																							
P ₂	3	0	2	9	0	2																																																																							
P ₃	2	1	1	2	2	2																																																																							
P ₄	0	0	2	4	3	3																																																																							
Module – 4																																																																													
Q.7	a.	What is paging? Explain with a neat diagram paging hardware with TLB.				10	L1 L2	CO4																																																																					
	b.	Explain the different strategies used to select a free hole from available holes.				05	L1	CO4																																																																					
	c.	What is Fragmentation? List and explain its types.				05	L2	CO4																																																																					
OR																																																																													
Q.8	a.	What is page fault? With a neat diagram explain steps in handling page fault.				08	L2	CO4																																																																					
	b.	Consider the page reference string for a memory with 3 frames determine the number of page faults using FIFO, optimal and LRU replacement algorithms. Which algorithms is more efficient? 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1				12	L3	CO4																																																																					
Module – 5																																																																													
Q.9	a.	Define File. List and explain different file operations and file attributes.				10	L1	CO5																																																																					
	b.	Explain the different file allocation methods.				10	L2	CO5																																																																					
OR																																																																													
Q.10	a.	What is Access Matrix? Explain the implementation of Access Matrix.				10	L2	CO5																																																																					
	b.	A drive has 5000 cylinders numbered 0 to 4999. The drive is currently servicing at a request 143 and previously served a request at 125. The queue of pending request in FIFO order. 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 starting from current head position. What is the total distance travelled (in cylinders) by a disk arm to satisfy the request using FCFS, SSTF, SCAN, LOOK and C-Look algorithm				10	L3	CO5																																																																					

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define Data Structures. Explain the classification of data structures with a neat diagram.	8	L2	CO1
	b.	Write a C Functions to implement pop , push and display operations for stacks using arrays.	7	L2	CO2
	c.	Differentiate structures and unions.	5	L2	CO1
OR					
Q.2	a.	Write an algorithm to evaluate a postfix expression and apply the same for the given postfix expression. 6 2 / 3 – 4 2 * +.	7	L3	CO2
	b.	Explain the dynamic memory allocation function in detail.	8	L2	CO1
	c.	What is Sparse matrix? Give the triplet form of a given matrix and find its transpose $A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$	5	L3	CO1
Module – 2					
Q.3	a.	Define Queue. Discuss how to represent a queue using dynamic arrays.	8	L2	CO2
	b.	Write a C Function to implement insertion () , deletion () and display () operations on circular queue.	6	L3	CO2
	c.	Write a note on Multiple stacks and queues with suitable diagram.	6	L2	CO2
OR					
Q.4	a.	What is a linked list? Explain the different types of linked list with neat diagram.	6	L2	CO3
	b.	Write a C function for the following on singly linked list with example : i) Insert a node of the beginning ii) Delete a node at the front iii) Display.	8	L3	CO3
	c.	Write the C function to add two polynomials.	6	L2	CO3

Module – 3					
Q.5	a.	Discuss how binary trees are represented using : i) Assay ii) Linked list.	6	L2	CO4
	b.	Define Threaded binary tree. Discuss In – threaded binary tree.	6	L2	CO4
	c.	Write the C function for the following additional list operation : i) Inverting Singly linked list ii) Concatenating Singly linked list.	8	L3	CO3
OR					
Q.6	a.	Discuss Inorder , Preorder , Postorder and Level order traversal with suitable function for each.	8	L3	CO4
	b.	Define the threaded binary tree. Construct threaded binary tree for the following element : A, B, C, D, E, F, G, H, I.	6	L2	CO4
	c.	Write a C function for the following : i) Insert a node at the beginning of doubly linked list. ii) Deleting a node at the end of the doubly linked list.	6	L3	CO3
Module – 4					
Q.7	a.	Define Forest , Transform the forest into a binary tree and traverse using inorder , preorder and postorder traversal with an example.	8	L1	CO5
	b.	Define Binary search tree. Construct a binary search tree for the following elements : 100 , 85 , 45 , 55 , 120 , 20 , 70 , 90 , 115 , 65 , 130 , 145.	6	L2	CO5
	c.	Discuss Selection tree with an example.	6	L2	CO5
OR					
Q.8	a.	Define Graph. Explain adjacency matrix and adjacency list representation with an example.	8	L2	CO5
	b.	Define the following terminology with example : i) Digraph ii) Weighted graph iii) Self loop iv) Connected graph.	6	L2	CO5
	c.	Briefly explain about Elementary graph operations.	6	L3	CO5
Module – 5					
Q.9	a.	Explain in detail about Static and Dynamic Hashing.	6	L2	CO5
	b.	What is Collision? What are the methods to resolve collision?	7	L2	CO5
	c.	Explain Priority queue with the help of an examples.	7	L2	CO5
OR					
Q.10	a.	Define Hashing. Explain different hashing functions with suitable examples.	12	L2	CO5
	b.	Write short note on : i) Leftist trees ii) Optimal binary search tree.	8	L3	CO5

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Object Oriented Programming with JAVA

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	List and explain any three features of object oriented programming.	6	L1	CO1
	b.	What do you mean by type conversion and type casting? Give examples.	8	L2	CO1
	c.	How to declare and initialize 1-D and 2-D arrays in Java. Give examples.	6	L2	CO1
OR					
Q.2	a.	List the short circuit operators and show the concept using few examples.	4	L2	CO1
	b.	With a java program, illustrate the use of ternary operator to find the greatest of three numbers.	6	L3	CO1
	c.	Develop a Java program to demonstrate the working of for each version of for loop. Initialize the 2D array with values and print them using for each.	10	L2	CO1
Module – 2					
Q.3	a.	Develop a program in Java to implement a stack of integers.	12	L3	CO2
	b.	What are constructors? Give the types and explain the properties of constructors. Support with appropriate examples.	8	L2	CO2
OR					
Q.4	a.	Illustrate with an example program to pass objects as arguments.	10	L2	CO2
	b.	Explain different access specifiers in Java with example program.	10	L2	CO2
Module – 3					
Q.5	a.	Define inheritance. List and explain different types of inheritance in Java with code snippets.	10	L2	CO3
	b.	Compare and contrast between overloading and overriding in Java with example program for each.	10	L2	CO3
OR					
Q.6	a.	Analyze an interface in Java and list out the speed of an interface. Illustrate with the help of a program the importance of an interface.	10	L2	CO3
	b.	List the different uses of final and demonstrate each with the of code snippets.	10	L2	CO3

1 of 2

Module – 4

Q.7	a.	Define a package. Explain how to create user defined package with example.	7	L2	CO4
	b.	Discuss about exception handling in Java. Give the framework of the exception handling block. List the types of exception.	8	L2	CO4
	c.	Develop a Java program to raise a custom exception for division by zero using try, catch, throw and finally.	5	L3	CO4

OR

Q.8	a.	Compare throw and throws keyword by providing suitable example program.	10	L2	CO4
	b.	Explain about the need for finally block.	5	L2	CO4
	c.	Discuss about chained exceptions.	5	L2	CO4

Module – 5

Q.9	a.	Define thread. Demonstrate creation of multiple threads with a program.	10	L2	CO5
	b.	Explain the two ways in which Java threads can be instantiated. Support your explanation with a sample program.	10	L2	CO5

OR

Q.10	a.	What is enumeration? Explain the methods values() and valueof().	10	L2	CO5
	b.	Explain about type wrappers and auto boxing.	10	L2	CO5

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain software process and software engineering practices.	10	L2	CO1
	b.	Explain the waterfall model and incremental model, with diagram.	10	L2	CO1
OR					
Q.2	a.	Explain Boehm Spiral process model with a neat diagram. Mention its advantages and disadvantages.	10	L2	CO1
	b.	Explain the five activities of a generic process framework for software engineering.	10	L2	CO1
Module – 2					
Q.3	a.	Explain the distinct tasks of requirement engineering.	10	L2	CO2
	b.	Illustrate the UML use case diagram for safe home system.	10	L2	CO2
OR					
Q.4	a.	Explain Class-Responsibility-Collaborator(CRC) modeling and data modeling with an example.	10	L2	CO2
	b.	Explain the elements of analysis model in requirement modeling.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the principles of agile process development.	10	L2	CO3
	b.	Explain the following : i) Adaptive software development ii) SCRUM	10	L2	CO3
OR					
Q.6	a.	Explain the concepts of extremes programming with a neat diagram.	10	L2	CO3
	b.	Explain design modeling principles that guide the respective framework activity.	10	L2	CO3
Module – 4					
Q.7	a.	Illustrate the project management life cycle with a neat diagram.	10	L2	CO4
	b.	Explain : i) Different ways of categorizing software projects ii) Smart objectives	10	L2	CO4
OR					
Q.8	a.	Explain the difference between traditional versus modern project management practices along with the role of management.	10	L3	CO4
	b.	Explain software development life cycle (ISO 12207) with a neat diagram.	10	L2	CO4
Module – 5					
Q.9	a.	Explain Quality Management System with principles of BS EN ISO-9001-2000.	10	L2	CO5
	b.	Explain the following : i) McCall model ii) Garvin's Quality Dimensions.	10	L2	CO5
OR					
Q.10	a.	Describe six generic functions allowed in automated estimation techniques of software projects.	10	L3	CO5
	b.	Explain COCOMO II model.	10	L2	CO5

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is data communication? List and explain characteristics and components of communication model.	06	L1	CO1
	b.	Define switching. Explain Circuit Switched Network and Packet Switched Network.	06	L2	CO1
	c.	With neat sketch, explain different layers of TCP/IP protocol suite.	08	L2	CO1
OR					
Q.2	a.	What are guided transmission media? Explain twisted pair cable in detail.	06	L1	CO1
	b.	What is Virtual Circuit Network (VCN)? With neat diagram, explain three phases involved in VCN.	08	L1	CO1
	c.	Write a note on Encapsulation and decapsulation at Source Host for TCP/IP protocol suite.	06	L2	CO1
Module – 2					
Q.3	a.	Define Redundancy. Explain CRC encoder and CRC decoder operation with block diagram.	08	L2	CO2
	b.	Distinguish between Flow Control and Error Control. Explain Stop and Wait Protocol.	08	L2	CO2
	c.	List and explain Control Fields of I-frames, S-frames and U-frames.	04	L2	CO2
OR					
Q.4	a.	What is Hamming distance? With example, explain Parity Check Code.	06	L1	CO2
	b.	Define Framing. Explain character oriented framing and bit-oriented framing.	06	L1	CO2
	c.	With flow diagram, explain CSMA/CA.	08	L2	CO2
Module – 3					
Q.5	a.	Explain virtual-circuit approach to route the packets in packet-switched network.	10	L2	CO3
	b.	Illustrate the working of OSPF and BGP.	10	L3	CO3
OR					
Q.6	a.	Explain IPv6 datagram format.	10	L2	CO3
	b.	Write an Dijkstra's algorithm to compute shortest path through graph.	06	L1	CO3
	c.	Write a note on Routing Information Protocol (RIP) algorithm.	04	L2	CO3
Module – 4					
Q.7	a.	Explain Go-Back-N protocol working.	10	L2	CO4
	b.	With neat sketch, explain three-way handshaking of TCP connection establishment.	10	L2	CO4

OR

Q.8	a.	With an outline, explain selective repeat protocol.	10	L2	CO4
	b.	List and explain various services provided by User Datagram Protocol (UDP).	10	L2	CO4

Module – 5

Q.9	a.	Briefly explain Secure Shell (SSH).	10	L2	CO4
	b.	Write a note on Request message and response message formats of HTTP.	10	L2	CO4

OR

Q.10	a.	With neat diagram, explain the basic model of FTP.	04	L2	CO4
	b.	Describe the architecture of electronic mail (e-mail).	06	L3	CO4
	c.	Briefly explain Recursive Resolution and Iterative Resolution in DNS.	10	L2	CO4

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Theory of Computation

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C														
Q.1	a.	Define the following with example : i) Language ii) String iii) Power of an alphabet.	3	L1	CO1															
	b.	Define DFA. Draw a DFA to accepts. i) The set of all strings that contain a substring aba. ii) To accept the strings of a's and b's that contain not more than three b's. iii) $L = \{w \in \{a, b\}^* : \text{No 2 consecutive characters are same in } w\}$.	10	L3	CO1															
	c.	Convert the following NFA to DFA. <table><tr><td></td><td>0</td><td>1</td></tr><tr><td>→ p</td><td>{p, q}</td><td>{p}</td></tr><tr><td>q</td><td>{r}</td><td>{r}</td></tr><tr><td>r</td><td>{s}</td><td>ϕ</td></tr><tr><td>* s</td><td>{s}</td><td>{s}</td></tr></table>		0	1	→ p	{p, q}	{p}	q	{r}	{r}	r	{s}	ϕ	* s	{s}	{s}	7	L2	CO1
	0	1																		
→ p	{p, q}	{p}																		
q	{r}	{r}																		
r	{s}	ϕ																		
* s	{s}	{s}																		
OR																				
Q.2	a.	Define the following with example : i) Alphabet ii) Reversal of string iii) Concatenation of Languages.	3	L1	CO1															
	b.	Design a DFA for the Language : $L = \{w \in \{0, 1\}^* : w \text{ is a string divisible by } 5\}$.	7	L3	CO1															
	c.	Define NFA. Obtain an ϵ - NFA which accepts strings consisting of 0 or more a's , followed by 0 or more b's followed by 0 or more C's. Also convert it to DFA.	10	L2	CO1															
Module – 2																				
Q.3	a.	Define Regular expression. Write the regular expression for the following languages : i) Strings of a's and b's starting with a and ending with b. ii) Set of strings that consists of alternating 0's and 1's. iii) $L = \{a^n b^m , (n + m) \text{ is even}\}$. iv) $L = \{w : w \bmod 3 = 0 , \text{ where } w \in \{a, b\}^*\}$.	10	L2	CO2															

	b.	Minimize the following finite automata using Table filling algorithm : <table><tr><th>δ</th><th>a</th><th>b</th></tr><tr><td>→ A</td><td>B</td><td>A</td></tr><tr><td>B</td><td>A</td><td>C</td></tr><tr><td>C</td><td>D</td><td>B</td></tr><tr><td>* D</td><td>D</td><td>A</td></tr><tr><td>E</td><td>D</td><td>F</td></tr><tr><td>F</td><td>G</td><td>E</td></tr><tr><td>G</td><td>F</td><td>G</td></tr><tr><td>H</td><td>G</td><td>D</td></tr></table>	δ	a	b	→ A	B	A	B	A	C	C	D	B	* D	D	A	E	D	F	F	G	E	G	F	G	H	G	D	10	L2	CO2
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H	G	D																														
OR																																
Q.4	a.	Construct ε - NFA for the following Regular expression : i) $(0 + 1) 0 1(1 + 0)$ ii) $1(0 + 1)^* 0$ iii) $(0 + 1)^* 0 1 1^*$	6	L1	CO2																											
	b.	Obtain the Regular expression that denotes the language accepted by Fig. Q4(b). Fig. Q4(b) Using Kleene's theorem.	6	L3	CO2																											
	c.	State the Pumping Lemma for the Regular Languages. And also prove that the following languages are not regular. i) $L = \{0^n 1^m \mid n \leq m\}$ ii) $L = \{0^n 1^m 2^n \mid n, m \geq 1\}$.	8	L1	CO2																											
Module – 3																																
Q.5	a.	Design CFG for the following languages : i) $L = \{a^n b^{n+3}, n \geq 0\}$ ii) $L = \{a^i b^j c^k, j = i + k, i \geq 0, k \geq 0\}$ iii) $L = \{w \mid w \bmod 3 > 0 \text{ where } w \in \{a\}^*\}$ iv) $L = \{a^m b^n \mid m \neq n\}$ v) Palindromes over 0 and 1.	10	L3	CO3																											
	b.	Consider the grammar G with productions. $S \rightarrow A b B / A / B$; $A \rightarrow aA / \varepsilon$; $B \rightarrow a B / b B / \varepsilon$. Obtain LMD , RMD and parse tree for the string aaabab. Is the given grammar ambiguous?	10	L2	CO3																											
OR																																
Q.6	a.	Define the following with example : i) Context free grammar ii) Left most Derivation iii) Parse tree iv) Ambiguous grammar.	4	L1	CO3																											
	b.	Design PDA for the language : $L = \{a^i b^j c^k \mid i + k = j, i \geq 0, k \geq 0\}$ and show the moves made by the PDA for the string aabbbc.	10	L3	CO3																											

	c.	Convert the following CFG's to PDA : $S \rightarrow a A$; $A \rightarrow a ABC / bB / a$; $B \rightarrow b$; $C \rightarrow c$.	6	L2	CO3
Module – 4					
Q.7	a.	Define CNF. Convert the following CFG to CNF $E \rightarrow E + T / T$ $T \rightarrow T * F / F$ $F \rightarrow (E) / I$ $I \rightarrow Ia / Ib / a / b$.	10	L2	CO4
	b.	Show that $L = \{0^n 1^n 2^n / n \geq 1\}$ is not context free.	4	L2	CO4
	c.	Prove that the family of context free languages is closed under union and concatenation.	6	L1	CO4
OR					
Q.8	a.	Define Greibach Normal Form. Convert the following CFG to GNF. $S \rightarrow AB$; $A \rightarrow aA / bB / b$; $B \rightarrow b$.	6	L2	CO4
	b.	Consider the following CFG : $S \rightarrow ABC / BaB$ $A \rightarrow aA / BaC / aaa$ $B \rightarrow bBb / a / D$ $C \rightarrow CA / AC$ $D \rightarrow \epsilon$ i) What are useless symbols? ii) Eliminate ϵ - productions , Unit productions and useless symbols from the grammar.	10	L3	CO4
	c.	Prove that the following languages are not context free. i) $L = \{a^i / i \text{ is prime}\}$ ii) $L = \{a^n / n \geq 1\}$.	4	L2	CO3
Module – 5					
Q.9	a.	Define a Turing machine and explain with neat diagram, the working of a basic Turing machine.	6	L1	CO4
	b.	Design a Turing machine to accept the language, $L = \{a^n b^n c^n / n \geq 1\}$. Draw the transition diagram and show the moves for the string aabbcc.	14	L4	CO4
OR					
Q.10	a.	Design a Turing machine to accept palindrome over $\{a, b\}$ and draw the transition diagram.	12	L4	CO5
	b.	Write a short notes on : i) Recursively Enumerable Language. ii) Multitape Turing Machine.	8	L1	CO5

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Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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1. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere
 2. The ratio between energy flows at different points in a food chain is known as
a) Ecological capacity b) Ecological efficiency
c) Ecological assimilation d) Ecological potential
 3. A predator is
a) An animal that is fed upon another animal
b) Animal that feeds upon both plants and animals
c) An animal that feeds upon another animal
d) A primary consumer
 4. Why Rann of Kutch attracts aquatic birds in monsoon season?
a) Because desert land is converted to forest land
b) Because desert land is converted to snow
c) Because desert land do not convert
d) Because desert land is converted to salt marshes
 5. Which kind of soil we can find on the surface of Thar desert?
a) Rocky b) Moist c) Fertile d) Aeolian
 6. Which of the following type of forest important for watersheds?
a) Tropical Evergreen forests b) Tropical Deciduous forests
c) Tropical Montana forests d) Grassland forest

7. Hot spots areas have
a) Low density of biodiversity
b) Only endangered plants
c) High density of hot springs
d) High density of biodiversity
8. Sustainable Development means
a) meeting present needs without compromising on future needs
b) progress of human beings
c) balance between human needs and the ability of earth to provide the resources
d) all of these
9. The term Alpha diversity refers to
a) Genetic diversity
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10. Algae , green plants and photosynthetic bacteria are
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11. Veld type grasslands are located at
a) South Africa
b) South America
c) Australia
d) Britain
12. Which pyramid is always upright?
a) Energy
b) Biomass
c) Numbers
d) Food chain
13. In what form is solar energy is radiated from the Sun?
a) Ultraviolet Radiation
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a) Magneto Hydro Dynamic
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15. The 'Miracle Material' that can turn CO₂ into liquid fuel is :
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19. What is a fuel cell?
a) Converts heat energy to chemical energy
b) Converts heat energy to electrical energy
c) Converts chemical energy to electrical energy
d) Converts kinetic energy to heat energy

20. Which one of the following is the apex organization in our country in the field of pollution control?
 a) Water Pollution Control Board b) State Pollution Control Board
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24. The international protocol to protect the Ozone layer is
 a) Vienna protocol b) Kyoto protocol
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25. Which is the best and the worst method of plume behavior for pollution dispersion?
 a) Lofting and fumigation b) Trapping and fanning
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26. What is called for a Temporary hearing loss?
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 c) 1.00 am to 7.00 am d) 10.00 pm to 6.00 am
28. In which section, if a person violates the noise pollution regulations, is liable for penalty according to Environmental Protection Act, 1986.
 a) Section 12 b) Section 15 c) Section 18 d) Section 19
29. 'Minamata Disease' is caused due to
 a) Lead b) Arsenic c) Mercury d) Cadmium
30. The process of reducing the fluoride content from water is called as
 a) Chlorination b) Fluoridation
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31. What is the Dissolved oxygen value required for the survival of aquatic species?
 a) 7 mg/L b) 8.2 mg/L c) 6.5 mg/L d) 4 mg/L
32. Which among the following is used to dump the waste collected in the cities?
 a) Land fills b) Ocean c) River d) All of these

45. Preparation of Guidelines for Environmentally sound Management of e – waste is a duty assigned to
a) Producer b) Consumer c) MOEFCC d) SP CB/PCC
46. What is India's global rank in e – waste?
a) 3 b) 13 c) 23 d) 33
47. When did the Karnataka State Pollution Control Board for prevention and control of water pollution constituted?
a) 1974 b) 1978 c) 1982 d) 1986
48. Aerosol consisting of liquid droplets is called as
a) Mist b) Dust c) Fog d) Aerosol
49. Which of the following is non – point source of water pollution?
a) Factories b) Sewage treatment plant
c) Urban and suburban land d) All of these
50. When is World Water day celebrated?
a) January 26th b) June 5th c) September 22nd d) March 22nd

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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1. What is Extended Producer Responsibility (EPR) as per the e – waste management rules in India?
 - a) The responsibility of consumer to manage e – waste
 - b) The responsibility of manufactures to manage e – waste throughout the product life cycle
 - c) The responsibility of retailers to manage e – waste disposal
 - d) The responsibility of informal recyclers to manage e – waste.
 2. Which international agreement regulates the transboundary movements of hazardous waste, including e – waste?
 - a) Kyoto Protocol
 - b) Paris agreement
 - c) Montreal Protocol
 - d) Basel convention
 3. Which colour bin is used for e – waste?
 - a) Blue
 - b) Green
 - c) Yellow
 - d) Black
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 - a) Lung cancer
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15. What is called for the process of burning municipal solid waste in a properly designed furnace under suitable temperature and operating conditions?
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20. Which of the following are the main contributors of the e –waste in the world?
 - I. Refrigerators / freezers , washing machines , dishwashers.
 - II. Small household appliances
 - III. Personal computers, telephones , laptops , printers.
 - IV. Gas cylinders, chimneys and home appliances
 - a) Only I, II, III
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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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II. Small household appliances
III. Personal computers, telephones , laptops , printers.
IV. Gas cylinders, chimneys and home appliances
a) Only I, II, III b) Only I & II c) Only I, III , IV d) All of these
31. Veld type grasslands are located at
a) South Africa b) South America c) Australia d) Britain
32. Which pyramid is always upright?
a) Energy b) Biomass c) Numbers d) Food chain
33. In what form is solar energy is radiated from the Sun?
a) Ultraviolet Radiation b) Infrared Radiation
c) Electromagnetic waves d) Transverse waves
34. What does MHD stands for in the energy field?
a) Magneto Hydro Dynamic b) Metal Hydrogen Detox
c) Micro Hybrid Drive d) Metering Head Differential
35. The ‘Miracle Material’ that can turn CO₂ into liquid fuel is :
a) Propane b) Copper c) Graphene d) Potassium

36. A tide whose difference between high and low tide is greatest.
a) Diurnal tide b) Neap tide c) Spring tide d) Ebb tide
37. Which of the turbine can be mounted vertically and horizontally.
a) Pelton wheel b) Kaplan turbine c) Gorlov turbine d) Francis turbine
38. Which type of fuel is removed from the reactor core after reaching end of core life service?
a) Burnt fuel b) Spent fuel c) Engine oil d) Radioactive fuel
39. What is a fuel cell?
a) Converts heat energy to chemical energy
b) Converts heat energy to electrical energy
c) Converts chemical energy to electrical energy
d) Converts kinetic energy to heat energy
40. Which one of the following is the apex organization in our country in the field of pollution control?
a) Water Pollution Control Board b) State Pollution Control Board
c) Central Pollution Control Board d) Air Pollution Control Board
41. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere
42. The ratio between energy flows at different points in a food chain is known as
a) Ecological capacity b) Ecological efficiency
c) Ecological assimilation d) Ecological potential
43. A predator is
a) An animal that is fed upon another animal
b) Animal that feeds upon both plants and animals
c) An animal that feeds upon another animal
d) A primary consumer
44. Why Rann of Kutch attracts aquatic birds in monsoon season?
a) Because desert land is converted to forest land
b) Because desert land is converted to snow
c) Because desert land do not convert
d) Because desert land is converted to salt marshes
45. Which kind of soil we can find on the surface of Thar desert?
a) Rocky b) Moist c) Fertile d) Aeolian
46. Which of the following type of forest important for watersheds?
a) Tropical Evergreen forests b) Tropical Deciduous forests
c) Tropical Montana forests d) Grassland forest
47. Hot spots areas have
a) Low density of biodiversity b) Only endangered plants
c) High density of hot springs d) High density of biodiversity

48. Sustainable Development means
- a) meeting present needs without compromising on future needs
 - b) progress of human beings
 - c) balance between human needs and the ability of earth to provide the resources
 - d) all of these
49. The term Alpha diversity refers to
- a) Genetic diversity
 - b) Community and ecosystem diversity
 - c) Species diversity within a community or ecosystem
 - d) Diversity among the plant
50. Algae , green plants and photosynthetic bacteria are
- a) Autotrophic
 - b) Heterotrophic
 - c) Decomposers
 - d) Consumers

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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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-
1. What is the Dissolved oxygen value required for the survival of aquatic species?
a) 7 mg/L b) 8.2 mg/L c) 6.5 mg/L d) 4 mg/L
 2. Which among the following is used to dump the waste collected in the cities?
a) Land fills b) Ocean c) River d) All of these
 3. Which type of waste includes items such as leftover food, fruit peels and yard trimmings?
a) Hazardous waste b) Organic waste
c) Bio – medical waste d) Electronic waste
 4. Which of the integrated waste management is reduced on an individual level?
a) Source Reduction b) Recycling c) Disposal d) Burning
 5. What is called for the process of burning municipal solid waste in a properly designed furnace under suitable temperature and operating conditions?
a) Landfill b) Recycling c) Vermicomposting d) Incineration
 6. The process of decomposition of biodegradable solid waste by earthworms is called
a) Landfill b) Vermicomposting c) Composting d) Shredding
 7. _____ is a liquid that passes through solid waste and extracts suspended impurities from it
a) Leachate b) Sludge c) Distilled water d) Municipal
 8. The colour code of plastic bag for disposing of microbial laboratory culture waste
a) black b) red c) blue d) white
 9. Average hospital waste produced per bed per day in Government hospital is
a) 1.5 to 2 kg b) 0.5 – 4 kg c) 0.5 to 1 kg d) 0.5 – 2 kg

10. Which of the following are the main contributors of the e –waste in the world?
 - I. Refrigerators / freezers , washing machines , dishwashers.
 - II. Small household appliances
 - III. Personal computers, telephones , laptops , printers.
 - IV. Gas cylinders, chimneys and home appliances
 - a) Only I, II, III b) Only I & II c) Only I, III , IV d) All of these
11. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
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20. Algae , green plants and photosynthetic bacteria are
 - a) Autotrophic b) Heterotrophic c) Decomposers d) Consumers

21. What is Extended Producer Responsibility (EPR) as per the e – waste management rules in India?
 - a) The responsibility of consumer to manage e – waste
 - b) The responsibility of manufactures to manage e – waste throughout the product life cycle
 - c) The responsibility of retailers to manage e – waste disposal
 - d) The responsibility of informal recyclers to manage e – waste.
22. Which international agreement regulates the transboundary movements of hazardous waste, including e – waste?
 - a) Kyoto Protocol
 - b) Paris agreement
 - c) Montreal Protocol
 - d) Basel convention
23. Which colour bin is used for e – waste?
 - a) Blue
 - b) Green
 - c) Yellow
 - d) Black
24. What are the health hazards which can be caused by E – waste?
 - a) Lung cancer
 - b) DNA damage
 - c) Brain
 - d) All of these
25. Preparation of Guidelines for Environmentally sound Management of e – waste is a duty assigned to
 - a) Producer
 - b) Consumer
 - c) MOEFCC
 - d) SP CB/PCC
26. What is India's global rank in e – waste?
 - a) 3
 - b) 13
 - c) 23
 - d) 33
27. When did the Karnataka State Pollution Control Board for prevention and control of water pollution constituted?
 - a) 1974
 - b) 1978
 - c) 1982
 - d) 1986
28. Aerosol consisting of liquid droplets is called as
 - a) Mist
 - b) Dust
 - c) Fog
 - d) Aerosol
29. Which of the following is non – point source of water pollution?
 - a) Factories
 - b) Sewage treatment plant
 - c) Urban and suburban land
 - d) All of these
30. When is World Water day celebrated?
 - a) January 26th
 - b) June 5th
 - c) September 22nd
 - d) March 22nd
31. _____ is caused by drinking water high in nitrates.
 - a) Cholera
 - b) Kidney problem
 - c) Liver problem
 - d) Methomoglobinemia
32. Bhopal gas tragedy took place in the year _____ and the gas responsible was _____.
 - a) 1964, Hydrogen fluoride
 - b) 1974, Methyl chloride
 - c) 1984, methyl ISO – cyanide
 - d) 1994, Methyl sulphate
33. The major chemical pollutants in photochemical smog are
 - a) NO, NO₂, VOC, O₃, PAN
 - b) N₂O, NO₂, VOC, O₃, PAN
 - c) NO, NO₂, VOC, O₂, PAN
 - d) NO, N₂O₅, VOC, O₃, PAN

48. Which type of fuel is removed from the reactor core after reaching end of core life service?
- a) Burnt fuel b) Spent fuel c) Engine oil d) Radioactive fuel
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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Artificial Intelligence

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define the following : i) Intelligence ii) Artificial Intelligence iii) Agent iv) Rationality v) Logical reasoning.	5	L2	CO1
	b.	Examine the AI literature to discover whether the following tasks can currently be solved by computers. i) Playing a decent game of table tennis (ping-pong) ii) Discovering and proving new mathematical theorems iii) Giving competent legal advice in a specialized area of law iv) Performing a complex a surgical operation.	8	L2	CO1
	c.	Implement a simple reflex agent for the vacuum environment. Run the environment with this agent for all possible initial dirt configurations and agent locations. Record the performance score for each configuration and the overall score.	7	L3	CO1
OR					
Q.2	a.	Is AI a science, or is it engineering or neither or both? Explain.	5	L2	CO1
	b.	Write pseudocode agent programs for the goal based and utility based agents.	8	L1	CO1
	c.	For each the following activities give a PEAS description. i) Playing a tennis match ii) Performing a high jump iii) Bidding on an item in an auction.	7	L1	CO1
Module – 2					
Q.3	a.	Explain why problem formulation must follow goal transformation.	5	L1	CO1
	b.	Give complete problem formulation for each of the following choose a formulation that is precise enough to be implemented. i) Using only four colors, you have to color a planar graph in such a way that no two adjacent regions have the same color. ii) A 3 – foot – tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get the bananas. The room contains two stackable, moveable, climbable 3-foot high crates.	8	L2	CO2
	c.	Prove each of the following statements or given counter example : i) Breadth – first search is a special case of uniform – cost search. ii) Uniform – cost search is a special case of A* search.	7	L2	CO2

OR					
Q.4	a.	Define the following terms with example. i) State space ii) Search node iii) Transition model iv) Branching factor.	8	L2	CO2
	b.	Show that the 8-puzzle states are divided in to two disjoint sets, such that any state is reachable from any other state in the same set, while no state is reachable from any state in the other set. Devise a procedure to decide which set a given state is in and explain why this is useful for generating random state.	7	L2	CO2
	c.	Describe a state space in which iterative deepening search performs much worse than depth first search for example, $O(n^2) \vee O(n)$.	5	L2	CO2
Module – 3					
Q.5	a.	Devise a state space in which A^* using GRAPH-SEARCH returns a suboptimal solution with $h(n)$ function that is admissible but inconsistent.	7	L2	CO3
	b.	Which of the following are correct? i) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) F(A \vee B)$ ii) $(A \vee B) \wedge (\neg C \vee \neg D \vee E) F(A \vee B) \wedge (\neg D \vee E)$ iii) $(A \vee B) \wedge \neg(A \Rightarrow B)$ is satisfiable iv) $(A \Leftrightarrow B) \Leftrightarrow C$ has the same number of models as $(A \Leftrightarrow B)$	8	L1	CO3
	c.	Consider a vocabulary with only four propositions, A, B, C and D. How many models are there for the following sentences? i) $B \vee C$ ii) $\neg A \vee \neg B \vee \neg C \vee \neg D$ iii) $(A \Rightarrow B) \wedge A \wedge \neg B \wedge C \wedge D$.	5	L1	CO3
OR					
Q.6	a.	Prove that if a heuristic is consistent, it must be admissible. Construct an admissible heuristic that is not consistent.	8	L1	CO3
	b.	Prove each of the following assertions : i) $\alpha \equiv \beta$ if and only if the sentence $(\alpha \Leftrightarrow \beta)$ is valid ii) $\alpha \neq \beta$ if and only if the sentence $\alpha \wedge \neg \beta$ is unsatisfiable.	7	L1	CO3
	c.	Prove, or find a counter example to each of the following assertions. i) If $\alpha \neq (\beta \wedge \gamma)$ then $\alpha \neq \beta$ and $\alpha \neq \gamma$ ii) If $\alpha \neq (\beta \vee \gamma)$ then $\alpha \neq \beta$ and $\alpha \neq \gamma$ (or) both	5	L1	CO3
Module – 4					
Q.7	a.	Which of the following are valid necessary true sentences? i) $(\exists x x = x) \Rightarrow (\forall y \exists z y = z)$ ii) $\forall x P(x) \vee \neg p(x)$ iii) $\forall x \text{ smart}(x) \vee (x = x)$	7	L1	CO4
	b.	Prove that universal Instantiation is sound that existential instantiation produces an inferentially equivalent knowledge base.	5	L1	CO4

	c.	Write down logical representations for the following sentences, suitable for use with generalized modulus ponens : i) Horses, cows and pigs are mammals ii) Bluebeard is Charlie's parent iii) Offspring and parent are inverse relations	8	L1	CO4
OR					
Q.8	a.	Consider a knowledge base containing just two sentence ; P(a) and P(b) does this knowledge base entail $\forall x P(x)$? Explain your answer in terms of models.	5	L2	CO4
	b.	Suppose a knowledge base contains just one sentence, $\exists x \text{AsHighAs}(x, \text{Everest})$ which of the following are legitimate results of applying existential instantiation? i) $\text{AsHighAs}(\text{Kilimanjaro}, \text{Everest})$ ii) $\text{AsHighAs}(\text{Kilimanjaro}, \text{Everest}) \wedge \text{AsHighAs}(\text{Benvevis}, \text{Everest})$	8	L2	CO4
	c.	Explain how to write any 3-SAT problem of arbitrary size using a single first order definite clause and no more than 30 ground facts.	7	L2	CO4
Module – 5					
Q.9	a.	i) Give a backward chaining proof of the sentence $7 \leq 3 + 9$. Show only the steps that leads to success ii) Give a forward chaining proof of the sentence $7 \leq 3 + 9$. Show only the steps that leads to success.	8	L1	CO5
	b.	Describe the differences and similarities between problem solving and planning.	5	L2	CO5
	c.	Prove that backward search with PDDL problems is complete.	7	L1	CO5
OR					
Q.10	a.	The following prolog code defines a predicate P $P(x, [x y]),$ $P(x, [y z]) :- P(x, z)$ i) Show proof trees and solutions for the queries $P(A, [2, 1, 3])$ and $P(z, [1, A, 3])$ ii) What standard list operation does P represent?	8	L1	CO5
	b.	Explain why dropping negative effects from every action schema in a planning problem results in a relaxed problems.	5	L2	CO5
	c.	Prove the following assertions about planning graphs : i) A literal that does not appear in the final level of the graph cannot be achieved. ii) The level cost of a literal in a serial graph is no greater than the actual cost of an optimal plan for achieving it.	7	L1	CO5

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Question Paper Version : A

Fifth Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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1. Which of the following is a Producer?
 a) Animals b) Human beings c) Plants d) Fish
 2. An ecosystem consists of
 a) Biotic species b) Biotic species and physical factors
 c) Only insects d) Only materials
 3. The term Ecology was coined by
 a) E.P. Odium b) A. G. Tansley c) Albert Einstein d) A.G. Haeckel
 4. Which year did the concept of sustainability first appear?
 a) 1992 b) 1978 c) 1980 d) 1987
 5. According to United Nations, the following are the significant issues of social progress.
 a) Instruction b) Public Health c) Living standards d) All of these
 6. The number of SDGS are
 a) 10 b) 15 c) 17 d) 16
 7. Which one of the following is not comes under SDG's?
 a) Zero Hunger b) Political activity c) Education d) Gender quality
 8. Which of the following has the largest population in food chain?
 a) Producers b) 1st degree consumers
 c) 2nd degree consumers d) Decomposers
 9. U.N era of scheduling for sustainable growth is from
 a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14

10. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake
11. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these
12. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
13. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
14. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear
15. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
16. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
17. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
18. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
19. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
20. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
21. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
22. The green house gas is
a) N₂O b) CH₄ c) CO₂ d) All of these
23. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
24. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
25. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB

26. What is the permissible range of p^H for drinking water as per the Indian standard?
 a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
27. Which of the following is a major cause of soil pollution?
 a) Accidents involving the vehicles that are transporting waste material.
 b) Pesticides and chemical fertilizers from agricultural lands
 c) Improper solid waste disposal d) All of these
28. What does E – Waste stand for?
 a) Environment Waste b) Electronic Waste
 c) Equipment Waste d) All of these
29. What is municipal solid waste disposal?
 a) The placement on the land b) The lack of waste
 c) In sanitary land fill site d) All of these
30. What is major cause of air pollution?
 a) Fossil fuel burning b) Industry emission
 c) Wild fire d) All of these
31. Average long – term weather of an area is called
 a) Weather conditions b) Seasonal variations
 c) Average weather d) Climate
32. Climate and global air circulations are mainly affected by the properties of
 a) Water and Air b) Temperature c) Precipitation d) None of these
33. The p^H of acid rain is
 a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2
34. Acid rain has been increasing day by day due to
 a) Urbanisation b) Industrialisation
 c) Increase in vehicle population d) None of these
35. Which is responsible for Ozone depletion?
 a) Methyl bromide b) CFC's
 c) Hydro chlorofluorocarbons d) All of these
36. The International protocol to protect the Ozone layer's
 a) Kytoto protocol b) Basal protocol c) Montreal protocol d) Viema protocol
37. The Ozone layer is locate upto _____ km above the earth surface.
 a) 1000 b) 50 c) 80 d) 100
38. Acid rain effects on
 a) Materials b) Plants c) Soil d) All of these
39. High – Concentration of fluoride content in drinking water causes.
 a) Dental flurosis b) Bone brittling c) Dental caries d) Plague
40. Desired concentration of fluoride content in drinking water in mg/ℓ is
 a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75

41. The First United National conference as Human Environment was held at Stickholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
42. To achieve the goal of clean environment, important strategies required are
a) Effective laws b) Active participation of the publics
c) Active participation of NGO's d) Both (a) and (b)
43. There are provisions for protection for our environment under
a) Indian Penal Code b) Police Act c) Municipal Act d) All of these
44. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
45. The Environmental Act of India was enacted in the year.
a) 1986 b) 1992 c) 1984 d) 1974
46. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these
47. Karnataka State "Pollution Control Board" was established in the year.
a) 1974 b) 1982 c) 1986 d) 1976
48. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
49. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
50. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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24. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear

25. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
26. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
27. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
28. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
29. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
30. Which of the following is conventional source of Energy?
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a) Instruction b) Public Health c) Living standards d) All of these
36. The number of SDGS are
a) 10 b) 15 c) 17 d) 16
37. Which one of the following is not comes under SDG's?
a) Zero Hunger b) Political activity c) Education d) Gender quality
38. Which of the following has the largest population in food chain?
a) Producers b) 1st degree consumers
c) 2nd degree consumers d) Decomposers
39. U.N era of scheduling for sustainable growth is from
a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14
40. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake

41. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
42. The green house gas is
a) N_2O b) CH_4 c) CO_2 d) All of these
43. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
44. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
45. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB
46. What is the permissible range of p^H for drinking water as per the Indian standard?
a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
47. Which of the following is a major cause of soil pollution?
a) Accidents involving the vehicles that are transporting waste material.
b) Pesticides and chemical fertilizers from agricultural lands
c) Improper solid waste disposal d) All of these
48. What does E – Waste stand for?
a) Environment Waste b) Electronic Waste
c) Equipment Waste d) All of these
49. What is municipal solid waste disposal?
a) The placement on the land b) The lack of waste
c) In sanitary land fill site d) All of these
50. What is major cause of air pollution?
a) Fossil fuel burning b) Industry emission
c) Wild fire d) All of these

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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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14. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
15. The Environmental Act of India was enacted in the year.
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17. Karnataka State "Pollution Control Board" was established in the year.
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18. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
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19. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
20. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron
21. Average long – term weather of an area is called
a) Weather conditions b) Seasonal variations
c) Average weather d) Climate
22. Climate and global air circulations are mainly affected by the properties of
a) Water and Air b) Temperature c) Precipitation d) None of these
23. The p^H of acid rain is
a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2

24. Acid rain has been increasing day by day due to
 - a) Urbanisation
 - b) Industrialisation
 - c) Increase in vehicle population
 - d) None of these
25. Which is responsible for Ozone depletion?
 - a) Methyl bromide
 - b) CFC's
 - c) Hydro chlorofluorocarbons
 - d) All of these
26. The International protocol to protect the Ozone layer's
 - a) Kytoto protocol
 - b) Basal protocol
 - c) Montreal protocol
 - d) Viema protocol
27. The Ozone layer is locate upto _____ km above the earth surface.
 - a) 1000
 - b) 50
 - c) 80
 - d) 100
28. Acid rain effects on
 - a) Materials
 - b) Plants
 - c) Soil
 - d) All of these
29. High – Concentration of fluoride content in drinking water causes.
 - a) Dental flurosis
 - b) Bone brittling
 - c) Dental caries
 - d) Plague
30. Desired concentration of fluoride content in drinking water in mg/ℓ is
 - a) 0.7 – 1.2
 - b) 0.5 – 2
 - c) 2 – 5
 - d) 1 – 1.75
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50. Which of the following is an example of Artificial Ecosystem?
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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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BMATEC301/BEC301/BBM301

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 AV Mathematics III for EC/ BM Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Statistical table and Mathematics formula handbook are allowed.
3. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1				M	L	C																	
Q.1	a.	Obtain the Fourier series of $f(x) = \frac{\pi - x}{2}$ in $0 < x < 2\pi$. Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.	6	L2	CO1																		
	b.	Find the Fourier series of $f(x) = x $ in $(-\ell, \ell)$. Hence show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$.	7	L3	CO1																		
	c.	Expand $f(x) = 2x - 1$ as a cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1																		
OR																							
Q.2	a.	Find the Fourier series of $f(x) = \begin{cases} 2x & \text{in } -\pi < x < 0 \\ 1 + \frac{\pi}{2} & \text{in } 0 < x < \pi \end{cases}$. Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$.	6	L2	CO1																		
	b.	Obtain the sine half range series of, $f(x) = \begin{cases} 1 - x & \text{in } 0 < x < \frac{1}{2} \\ 4 & \text{in } \frac{1}{2} < x < 1 \end{cases}$	7	L2	CO1																		
	c.	Determine the constant term and the first cosine and sine terms of the Fourier series expansion of y from the following data : <table><tr><td>x°:</td><td>0</td><td>45</td><td>90</td><td>135</td><td>180</td><td>225</td><td>270</td><td>315</td></tr><tr><td>y:</td><td>2</td><td>$\frac{3}{2}$</td><td>1</td><td>$\frac{1}{2}$</td><td>0</td><td>$\frac{1}{2}$</td><td>1</td><td>$\frac{3}{2}$</td></tr></table>	x°:	0	45	90	135	180	225	270	315	y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	7	L1	CO1
x°:	0	45	90	135	180	225	270	315															
y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$															
Module – 2																							
Q.3	a.	Find the Fourier transform of the function, $f(x) = \begin{cases} 1 & \text{for } x \leq a \\ 0 & \text{for } x > a \end{cases}$. Hence evaluate $\int_0^\infty \frac{\sin x}{x} dx$.	6	L2	CO2																		
	b.	Find the Fourier sine and cosine transforms of $f(x) = e^{-\alpha x}$, $\alpha > 0$.	7	L2	CO2																		
	c.	Find the Fourier sine transform of $\frac{e^{-ax}}{x}$, $a > 0$.	7	L3	CO2																		

OR																		
Q.4	a.	If $f(x) = \begin{cases} 1-x^2, & x < 1 \\ 0, & x \geq 1 \end{cases}$, find the Fourier transform of $f(x)$ and hence find the value of, $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx$.	6	L2	CO2													
	b.	Find the Fourier sine transform of $f(x) = e^{- x }$ and hence evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx, m > 0$.	7	L3	CO2													
	c.	Find the Discrete fast fourier of signal $= (0, 1, 49)^T$	7	L3	CO2													
Module – 3																		
Q.5	a.	Find the z-transform of, (i) $\cosh n\theta$ (ii) $\sinh n\theta$	6	L1	CO3													
	b.	If $V(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$, evaluate u_0, u_1 and u_2	7	L2	CO3													
	c.	Find the inverse z-transform of, $\frac{z}{(z-1)(z-2)}$.	7	L2	CO3													
OR																		
Q.6	a.	Solve by using z-transforms, $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = 0 = y_1$	6	L3	CO3													
	b.	Find $z^{-1} \left[\frac{5z}{(3z-1)(2-z)} \right]$.	7	L2	CO3													
	c.	Solve by using z-transforms $u_{n+2} - 5u_{n+1} + 6u_n = z_n$ with $u_0 = 0 = u_1$.	7	L3	CO3													
Module – 4																		
Q.7	a.	Solve $\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} + 6y = 0$.	6	L1	CO4													
	b.	Solve $(D^2 + 1)y = x^2 + 4x - 6$.	7	L2	CO4													
	c.	Using the method of variation of Parameters of $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}$	7	L3	CO4													
OR																		
Q.8	a.	Solve $6\frac{d^2y}{dx^2} + 17\frac{dy}{dx} + 12y = e^{-x}$.	6	L2	CO4													
	b.	Solve the Cauchy's differential equation, $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + 8y = 65\cos(\log x)$.	7	L2	CO4													
	c.	The charge q in a series circuit containing an Inductance L , Capacitance C , emf E satisfy the differential equation, $L\frac{d^2q}{dt^2} + \frac{q}{C} = E$. Express q in terms of t .	7	L3	CO4													
Module – 5																		
Q.9	a.	Fit a second degree parabola $y = a + bx + cx^2$ into least square sense for the data and estimate y at $x = 6$. <table><tr><td>x:</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y:</td><td>10</td><td>12</td><td>13</td><td>16</td><td>19</td></tr></table>	x:	1	2	3	4	5	y:	10	12	13	16	19	6	L1	CO5	
x:	1	2	3	4	5													
y:	10	12	13	16	19													

	b.	Find a correlation coefficient for the two variables x and y.	7	L2	CO5																						
		<table><tr><td>x:</td><td>92</td><td>89</td><td>87</td><td>86</td><td>83</td><td>77</td><td>71</td><td>63</td><td>53</td><td>50</td></tr><tr><td>y:</td><td>86</td><td>83</td><td>91</td><td>77</td><td>68</td><td>85</td><td>52</td><td>82</td><td>37</td><td>57</td></tr></table>	x:	92	89	87	86	83	77	71	63	53	50	y:	86	83	91	77	68	85	52	82	37	57			
x:	92	89	87	86	83	77	71	63	53	50																	
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	c.	Ten students got the following percentage of marks in two subjects x and y. Compute the rank correlation coefficient.	7	L2	CO5																						
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y:	84	51	91	60	68	62	86	58	53	47																	
OR																											
Q.10	a.	If θ is the angle between the lines of regression show that $\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right).$	6	L2	CO5																						
	b.	Obtain the lines of regression and hence find the coefficient of correlation for the data, <table><tr><td>x:</td><td>1</td><td>3</td><td>4</td><td>2</td><td>5</td><td>8</td><td>9</td><td>10</td><td>13</td><td>15</td></tr><tr><td>y:</td><td>8</td><td>6</td><td>10</td><td>8</td><td>12</td><td>16</td><td>16</td><td>10</td><td>32</td><td>32</td></tr></table>	x:	1	3	4	2	5	8	9	10	13	15	y:	8	6	10	8	12	16	16	10	32	32	7	L2	CO5
x:	1	3	4	2	5	8	9	10	13	15																	
y:	8	6	10	8	12	16	16	10	32	32																	
	c.	If $8x - 10y + 66 = 0$ and $40x - 18y = 214$ are the two regression lines. Find $\bar{x}$, $\bar{y}$ and r. Find σ_y if $\sigma_x = 3$.	7	L2	CO5																						

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Identify the meaning of Research and brief out the objective and motivation in engineering research.	10	L1	CO1
	b.	Explain brief about research cycle and verify with the research flow diagram.	10	L1	CO1
OR					
Q.2	a.	Identify the types of engineering research and briefly explain them.	10	L1	CO1
	b.	Explain about the different types of research misconduct.	10	L1	CO1
Module – 2					
Q.3	a.	Explain about the importance of literature review and technical reading.	10	L2	CO2
	b.	Mention the various benefits of bibliographic databases.	10	L1	CO2
OR					
Q.4	a.	Identify the impact of technical reaction and brief about it.	10	L1	CO2
	b.	Enumerate the impact of title and keywords on citation with example.	10	L2	CO2
Module – 3					
Q.5	a.	Define Intellectual properties and explain about its types.	10	L1	CO3
	b.	Explain about the key aspect of patent law.	10	L2	CO3
OR					
Q.6	a.	Explain about the assessment of novelty.	10	L1	CO3
	b.	Brief about the patent procedure in India.	10	L1	CO4
Module – 4					
Q.7	a.	Mention and brief about the justification for copyright law.	10	L2	CO4
	b.	Explain about the basic concepts of under lying copyright law.	10	L1	CO4
OR					
Q.8	a.	Brief about the various representations of sound recordings.	10	L2	CO5
	b.	Explain about TRIPS agreement in detail.	10	L1	CO5

Module – 5					
Q.9	a.	Explain about the justification of protection designs.	10	L2	CO5
	b.	Brief about the excluded subjected matter in the context of design protection.	10	L1	CO5
OR					
Q.10	a.	What are the rights of the owner of designs? Explain.	10	L1	CO5
	b.	Brief about the Assignment of Design Rights.	10	L1	CO5

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Automata Theory and Compiler Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

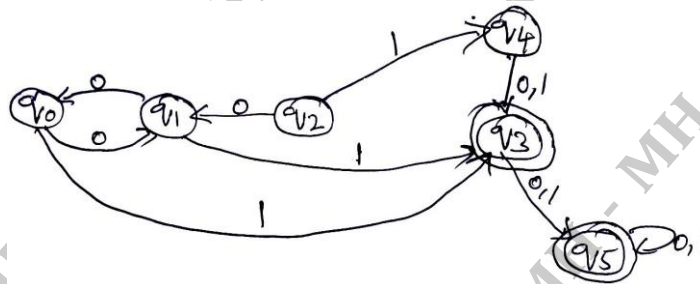
Module-1

- 1 a. Define the three basic concepts of Automata. Also construct a DFA that accepts all strings that have the first and last letter different on $\Sigma = \{a, b\}$. Justify the DFA with example. (10 Marks)
- b. Solve by converting the following NFA to DFA. (10 Marks)



OR

- 2 a. Explain the different phases of a compiler with neat block diagram and convert the source code. Position = Initial + rate * 60 into target code. (10 Marks)
- b. Solve by Minimizing the following DFA. (10 Marks)



Module-2

- 3 a. Define the formal definition of Regular expression. Also write the regular expression for the following : i) Set of strings consisting of Even numbers of 'a' s followed by odd number of 'b' s on $\Sigma = \{a, b\}$.
 ii) $L = \{a^n b^m : (n + m) \text{ is even}\}$.
 iii) $L = \{a^n b^m : n \geq 4, m \leq 3\}$. Justify the answer. (10 Marks)
- b. Explain Input buffering in Lexical Analyzer. Define Token , Patterns and Lexemes with examples. Also write the tokens for $E = m * c * 2$. (10 Marks)

OR

- 4 a. Define Regular Definitions. Write the Regular Definitions for 'C' identifiers and unsigned numbers using short hands notations and write the transition diagram. (10 Marks)
- b. State and prove pumping lemma theorem for Regular languages. (10 Marks)

Module-3

- 5 a. Define Context free grammar. Write a CFG for the following :
 i) To generate strings of palindrome over $\Sigma = \{0, 1\}$.
 ii) $L = \{a^i b^j \mid i \neq j, i \geq 0 \text{ and } j \geq 0\}$
 iii) $L = \{0^m 1^n 2^n \mid m \geq 1, n \geq 0\}$. Justify the answer. (10 Marks)

- b. Define Left recursion and left factoring. Also remove the left recursion and left factoring for the Grammar
- $$E \rightarrow E + T \mid T$$
- $$T \rightarrow id \mid id [] \mid id [X]$$
- $$X \rightarrow E, E \mid E.$$

(10 Marks)

OR

- 6 a. Define Ambiguous grammar. Show that the following is ambiguous. (10 Marks)

$S \rightarrow i c t s \mid i c t s e s \mid a$
 $G \rightarrow b$ for the string ibtibtaea

- b. Consider the grammar

$$E \rightarrow T E'$$

$$E' \rightarrow + T E' \mid \epsilon$$

$$T \rightarrow F T'$$

$$T' \rightarrow * F T' \mid \epsilon$$

$$F \rightarrow (E) \mid id$$

- i) Compute FIRST and Follow sets.
 ii) Using FIRST and Follow sets construct the Predictive LL (1) parsing table. (10 Marks)

Module-4

- 7 a. Define Non Deterministic Pushdown Automata. Construct an NPDA for the Language $L = \{W \in (a, b)^* : n_a(w) = n_b(w)\}$ and draw the transition diagram. (10 Marks)
- b. Define Handle and Handle Pruning. For the following grammar perform shift reduce for the string $id_1 + id_2 * id_3$.
- $$E \rightarrow E + E$$
- $$E \rightarrow E * E$$
- $$E \rightarrow (E)$$
- $$E \rightarrow id.$$

(10 Marks)

OR

- 8 a. Define Instantaneous Description in Pushdown Automata. Construct an NPDA for the Language $L = \{WCW^R : W \in (a, b)^*\}$. (10 Marks)
- b. Consider the Grammar.
- $$S \rightarrow L = R \mid R$$
- $$L \rightarrow * R \mid id$$
- $$R \rightarrow L$$

Verify the grammar is SLR (1) or not through the suitable parsing table. (10 Marks)

Module-5

- 9 a. Define Turing Machine. Construct a Turing Machine to recognize the Language. $L = \{a^n b^n : W \in \{a, b\}^* n \geq 1\}$. (10 Marks)
- b. Write the SDD for the grammar. Also construct the Annotated Parse tree for $5 * 6 + 7$;
- $$S \rightarrow EN$$
- $$E \rightarrow E + T$$
- $$E \rightarrow E - T$$
- $$E \rightarrow T$$
- $$T \rightarrow T * F$$
- $$T \rightarrow T / F$$
- $$T \rightarrow F$$
- $$F \rightarrow (E)$$
- $$F \rightarrow digit$$
- $$N \rightarrow ;$$

(10 Marks)

OR

- 10 a. Construct a Turing Machine to recognize the Language.
 $L = \{0^n 1^n 2^n \mid n \geq 1\}$ and trace the string 0 0 1 1 2 2. (12 Marks)
- b. For the Grammar construct the SDD and the annotated parse tree for the string $3 * 5 * 4$ and show the Evaluation order.
- $T \rightarrow FT'$
 $T' \rightarrow * FT'$
 $T' \rightarrow \epsilon$
 $F \rightarrow \text{digit}.$ (08 Marks)

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss OSI Reference model with a neat diagram. (10 Marks)
- b. Explain Unicast, Multicast and broadcast in computer networks. (10 Marks)

OR

- 2 a. Discuss the following transmission medium with diagram,
 - (i) Co-axial cable.
 - (ii) Fibre optic cable.(10 Marks)
- b. Explain the design issues in computer networks. (10 Marks)

Module-2

- 3 a. Explain error detecting codes and obtain the CRC code for the frame given polynomial 1101011111 using the generator $G(x) = x^4 + x + 1$. (10 Marks)
- b. Explain framing with Byte Count and Flag bits with bit stuffing. (10 Marks)

OR

- 4 a. Illustrate the Hamming code method with an example. (10 Marks)
- b. Explain the following with examples:
 - (i) Binary convolution code.
 - (ii) Reed Solomen code.(10 Marks)

Module-3

- 5 a. Discuss Store and Forward packet switching. (10 Marks)
- b. Explain the services provided by Network layer to Transport layer. (10 Marks)

OR

- 6 a. Discuss shortest path algorithm. (10 Marks)
- b. Explain the approaches to congestion control in Network layer. (10 Marks)

Module-4

- 7 a. Explain Berkeley Sockets in detail. (10 Marks)
- b. Explain socket programming with an example. (10 Marks)

OR

- 8 a. Explain TCP protocol with TCP segment header. (10 Marks)
b. Explain TCP connection establishment and TCP connection release with code snippet. (10 Marks)

Module-5

- 9 a. Explain the process communication in the Application layer. (10 Marks)
b. Discuss the Transport services provided by the Internet. (10 Marks)

OR

- 10 a. Explain the web and HTTP with Request response behaviour. (10 Marks)
b. Discuss the Electronic Mail in the Internet. (10 Marks)

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21CS53

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Database Management Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain in detail the characteristics of database approach. (08 Marks)
- b. Define the following terms and also give example :
i) Database ii) DBMS (04 Marks)
- c. List and explain the advantages of using DBMS Approach. (08 Marks)

OR

- 2 a. Explain cardinality ratio and participation constraints along with an example. (06 Marks)
- b. With a neat diagram explain the three schema architecture. (06 Marks)
- c. Draw an ER diagram for library database by considering at least 5 entities. (08 Marks)

Module-2

- 3 a. Explain in detail characteristics of Relations. (06 Marks)
- b. Discuss different types of update operations on relational database. Also give an example. (06 Marks)
- c. Write a note on Natural join and division operation. (08 Marks)

OR

- 4 a. Consider the 2 tables. Show the result of the following :

R ₁		
a ₁	a ₂	a ₃
20	L	15
15	m	18
25	L	16

R ₂		
b ₁	b ₂	b ₃
20	L	6
25	n	8
28	l	4

(i) $R_1 \bowtie R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

(ii) $R_1 \Join R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

(iii) $R_1 \bowtie R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

(iv) $R_1 \Join R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

- b. With an example explain steps of ER to Relational Mapping algorithm.

(08 Marks)

(12 Marks)

Module-3

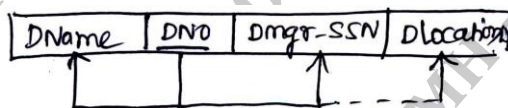
- 5 a. For the following Database schema.
 Employee (Fname, Minit, Lname, SSN, Bdate, Address, Salary, SuperSSN, DNo)
 Department(DName, Dno, Mgr_SSN, Mgr_Startdate)
 Dept_Locations(Dno, Dlocation)
 Project(PName, Prj_no, Plocation, Dnum)
 Works_on(ESSN, Prj_no, Hours)
 Dependent(ESSN, DependentName, Sex, Bdate, Relationship)
 Write SQL Queries for the following :
- Find sum_of_salaries of all employees who work in Dept No 10, average salaries of all employees who work in Dept No 10.
 - List all employees who do not have any dependent.
 - For each project, retrieve the project number and the number of employees who work on that project.
 - Make list of all project numbers for projects that involve an employee whose last name is 'Kumar'. (08 Marks)
- b. Write command that is used for table creation. Explain how primary key, foreign key are specified in SQL during table creation with suitable example. (06 Marks)
- c. Explain view in SQL, with suitable example. (06 Marks)

OR

- 6 a. Explain stored procedures in SQL with example. (06 Marks)
- b. How triggers are defined in SQL? Explain with an example. (06 Marks)
- c. Write a note on : (i) Cursor (ii) Assertions (08 Marks)

Module-4

- 7 a. List and explain the informal Design guidelines for relation schemas. (08 Marks)
- b. Define the following :
 (i) Functional dependency (ii) Key (iii) Superkey (iv) Prime attribute (06 Marks)
- c. For the given schema, discuss the 3 main techniques to achieve first normal form.



(06 Marks)

OR

- 8 a. Explain in detail 2nd Normal form and 3rd Normal form along with example. (08 Marks)
- b. Write an algorithm for determining X^+ , the closure of X under F. Give an example. (06 Marks)
- c. Write a note on 4th Normal form. (06 Marks)

Module-5

- 9 a. Define Transaction. Discuss ACID properties. (06 Marks)
- b. With neat diagram explain transition diagram of a transaction. (06 Marks)
- c. Explain the Lost Update problem and Temporary update problem with respect to concurrent transaction execution. (08 Marks)

OR

- 10 a. Briefly discuss 2-phase locking techniques for concurrency control. (10 Marks)
- b. Write a note on :
 i) Deadlock prevention protocols ii) Basic Timestamp ordering algorithm (10 Marks)

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21CS71

Seventh Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025

Big Data Analytics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the evolution of Big Data. (06 Marks)
- b. Explain the characteristics of Big Data. (04 Marks)
- c. Explain Data Architecture Design, with a neat diagram. (10 Marks)

OR

- 2 a. Explain Analytics Scalability to Big Data and Massive parallel processing platforms. (12 Marks)
- b. Explain Big Data Analytics applications with one case study. (08 Marks)

Module-2

- 3 a. List and explain the core components of Hadoop. (10 Marks)
- b. Explain Hadoop Distributed File System. (10 Marks)

OR

- 4 a. Define MapReduce Frame work and its functions. (06 Marks)
- b. Explain steps on the request to MapReduce and the types of process in MapReduce. (10 Marks)
- c. Explain in brief on Flume Hadoop Tool. (04 Marks)

Module-3

- 5 a. Explain about No SQL datastore and its characteristics. (10 Marks)
- b. Describe the principle of working of the CAP theorem. (10 Marks)

OR

- 6 a. Demonstrate the working of key-value store with an example. (10 Marks)
- b. Describe the features of MongoDB, and its industrial application. (10 Marks)

Module-4

- 7 a. Explain the process in MapReduce when client submitting a job, with a neat diagram. (10 Marks)
- b. Explain Hive Integration and workflow steps involved with a diagram. (10 Marks)

OR

- 8 a. Using HiveQL for the following :
i) Create a table with partition
ii) Add, rename and drop a partition to a table. (10 Marks)
b. What is PIG in BigData? Explain the feature of PIG. (10 Marks)

Module-5

- 9 a. Explain linear and non-linear relationship with essential graphs in machine learning. (10 Marks)
b. Write the block diagram of text mining process and explain its phases. (10 Marks)

OR

- 10 a. With a neat diagram, write the steps in K-means clustering. (10 Marks)
b. Explain the purpose of web usage analytics and the significance of web graphs. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Cloud Computing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat diagram, explain cloud computing and its historical development. (10 Marks)
- b. List the characteristics and benefits of cloud computing. (10 Marks)

OR

- 2 a. Explain in brief the services provided by the following cloud service provider:
 - i) Amazon web service
 - ii) Microsoft azure
 - iii) Google AppEngine. (10 Marks)
- b. Write a note on challenges in cloud computing. (10 Marks)

Module-2

- 3 a. Define virtualization and explain hardware level virtualization with its advantages. (10 Marks)
- b. Discuss the taxonomy of virtualization techniques at different levels. (10 Marks)

OR

- 4 a. What are the characteristics of virtualized environment? (10 Marks)
- b. Explain with a neat diagram Type-I and Type-II hypervisor. (10 Marks)

Module-3

- 5 a. Explain the different types of cloud. (10 Marks)
- b. What is IaaS? Explain its reference implementation with a neat diagram. (10 Marks)

OR

- 6 a. Explain the economics of the cloud. (10 Marks)
- b. What does the acronym SaaS mean? How does it relate to cloud computing? (10 Marks)

Module-4

- 7 a. Analyze the various cloud security risks that organization face when utilizing cloud computing services. (10 Marks)
- b. Explain the security risks posed by a management OS. (10 Marks)

OR

- 8 a. Discuss the traditional concept of trust and trust necessary for online activities. (10 Marks)
- b. Explain in detail virtual machine security. (10 Marks)

Module-5

- 9 a. Describe Amazon EC2 and its basic features. (10 Marks)
- b. Analyze how cloud computing technology can be applied to support remote ECG monitoring. (10 Marks)

OR

- 10 a. What is a bucket? What type of storage does it provide? (10 Marks)
- b. Examine the core components of AppEngine. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8 = 50, will be treated as malpractice.

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Internet of Things

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1.
 - a. In the context of Evolution of IoT, with neat diagram, illustrate the sequence of technological developments leading to the shaping of the modern day IoT. (06 Marks)
 - b. With diagram, explain enabling IoT and the complex Interdependence of technologies. (08 Marks)
 - c. Differentiate between the following :
 - (i) IoT versus M2M.
 - (ii) IoT versus CPS.
 - (iii) IoT versus WoT. (06 Marks)

OR

2.
 - a. With respect to the IoT networking components, define the following :
 - (i) IoT NODE.
 - (ii) IoT Router.
 - (iii) IoT LAN
 - (iv) IoT Gateway
 - (v) IoT Proxy (10 Marks)
 - b. Discuss the following addressing strategies in IoT:
 - (i) Address Management Classes.
 - (ii) Addressing during node Mobility. (10 Marks)

Module-2

3.
 - a. Define sensors and with diagram, outline the simple sensing operation. (04 Marks)
 - b. Discuss Scalar and Vector sensors and draw the functional blocks of a typical sensor node in IoT. (06 Marks)
 - c. With neat diagram, explain the different sensing types commonly encountered in IoT. (10 Marks)

OR

4.
 - a. Define Actuator and with diagram, discuss the outline of a simple actuation mechanism. (04 Marks)
 - b. Explain the various actuators classes any 5 in IoT. (10 Marks)
 - c. Discuss actuator characteristics that define all actuators. (06 Marks)

Module-3

5.
 - a. List and discuss common data types used in IoT applications. (06 Marks)
 - b. Explain the various processing topologies in IoT with necessary diagrams. (08 Marks)
 - c. Illustrate the importance of processing in IoT. (06 Marks)

OR

- 6 a. With neat diagram, explain the processing offloading paradigm for the development of IoT-based solutions. (10 Marks)
- b. Determine the importance of choosing the right processing topologies and associated considerations while designing IoT applications. (10 Marks)

Module-4

- 7 a. List common connectivity protocols in IoT. (04 Marks)
- b. Explain the salient features and application scope of any 5 connectivity protocols. (10 Marks)
- c. Differentiate between Wi-Fi and Bluetooth connectivity protocols in IoT. (06 Marks)

OR

- 8 a. With necessary diagrams, explain in detail any four connectivity protocols in IoT. (10 Marks)
- b. Determine the requirements associated with any of IoT connectivity protocols in real-world solutions. (10 Marks)

Module-5

- 9 a. Describe in detail, the various Infrastructure protocols in IoT-based communication technologies. (10 Marks)
- b. Explain the following discovery protocols :
- (i) Physical Web
 - (ii) mDNS
 - (iii) Universal plug and play (UPnP). (10 Marks)

OR

- 10 a. With neat diagram, discuss the illustration of the various facets of the Interoperability in IoT. (10 Marks)
- b. Discuss any four IoT Interoperability standards. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Software Architecture and Design Patterns

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is design pattern? Explain essential elements of design patterns. (06 Marks)
- b. Explain how the design pattern catalog is organized. (06 Marks)
- c. Construct and explain the sequence diagram for renewing books in Library system. (08 Marks)

OR

- 2 a. Illustrate how the design patterns and frameworks are different. (06 Marks)
- b. Discuss the step-by-step approach of applying a design pattern effectively with relevant examples. (06 Marks)
- c. Explain the common causes of redesign and its design patterns for addressing them. (08 Marks)

Module-2

- 3 a. Describe the intent, motivation and applicability of adapter design pattern. (08 Marks)
- b. Explain the intent, applicability, structure, participants and collaboration sections of proxy pattern. (12 Marks)

OR

- 4 a. Discuss the consequences and implementation issues of bridge pattern. (08 Marks)
- b. Give the structure of composite pattern. Explain the issues to be considered when implementing the composite design pattern. (12 Marks)

Module-3

- 5 a. Define behavioural patterns. Explain motivation applicability and structure of chain of responsibility. (10 Marks)
- b. Explain when to use momento, observer, state, command and mediator design patterns. (10 Marks)

OR

- 6 a. Discuss the implementation issues of observer pattern. (10 Marks)
- b. Explain motivation, structure and implementation of iterator design pattern. (10 Marks)

Module-4

- 7 a. With a neat diagram, explain MVC architecture and it's alternative view. (08 Marks)
- b. What are the benefits of MVC architectural pattern? (06 Marks)
- c. Explain use case for drawing a line. (06 Marks)

OR

- 8 a. Illustrate the sequence of operations involved in drawing a line with a neat diagram. (08 Marks)
b. Explain implementing the undo operation. (06 Marks)
c. Explain the characteristics of architectural patterns in pattern based solutions. (06 Marks)

Module-5

- 9 a. With a neat diagram, explain the basic architecture of client / server systems. (08 Marks)
b. Write short notes on GET and POST methods. (06 Marks)
c. Explain the state transition diagram for issuing a book. (06 Marks)

OR

- 10 a. Describe how the library system can be deployed on the World Wide Web. (08 Marks)
b. Explain the different steps to setup remote object system. (08 Marks)
c. Draw and explain directory structure for Servlet. (04 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Deep Learning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the concept of tasks(T), Performance (P) and Experience (E). Describe the following with respect to tasks performance and experience.
 - i) Checker learning problem
 - ii) Handwriting recognition learning problem (12 Marks)
- b. Explain the concept of supervised and unsupervised learning with example. (08 Marks)

OR

- 2 a. Explain the historical trends in deep learning. (10 Marks)
- b. Define supervised and unsupervised learning algorithm. Describe KNN and K means algorithm. (10 Marks)

Module-2

- 3 a. Explain about gradient based learning. (10 Marks)
- b. Explain the concept of Back propagation and how it helps in a Neural network. (10 Marks)

OR

- 4 a. Define Regularization. Describe L^1 and L^2 regularization. (10 Marks)
- b. Given $W = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $W = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $b = 0$ draw feed forward network and evaluate XOR function. (10 Marks)

Module-3

- 5 a. What are the challenges in neural network optimization? (10 Marks)
- b. Explain the following algorithms
 - i) RMSProp
 - ii) RMSProp with momentum. (10 Marks)

OR

- 6 a. Explain stochastic gradient descent and momentum algorithms (10 Marks)

- b. Give the list of adaptive learning rates algorithms. Write the Ada Grad algorithm. (10 Marks)

Module-4

- 7 a. Explain the following with suitable diagram.
i) Sparse interactions ii) Parameter sharing. (10 Marks)
- b. Explain briefly variant of the CNN models. (10 Marks)

OR

- 8 a. Differentiate locally connected layers, tiled convolution and standard convolution with suitable example and diagram. (10 Marks)
- b. Explain the different layers in CNN models and its function with a neat diagram. (10 Marks)

Module-5

- 9 a. Discuss about Bidirectional Recurrent neural networks. (10 Marks)
- b. What is speech recognition? Explain the different types of speech recognition systems. (10 Marks)

OR

- 10 a. Explain Long Short-Term Memory (LSTM) working principles along with all the equations. (10 Marks)
- b. What is Natural language processing? Explain different steps involved in NLP. (10 Marks)

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21CS752

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Introduction to AI and ML

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Artificial Intelligence. Explain the below approaches to Artificial Intelligence with respective definitions.
i) Acting Humanly ii) Thinking Humanly (10 Marks)
b. Define Agent. With a neat diagram, explain how agents interact with environment through sensors and actuators. (10 Marks)

OR

- 2 a. Explain the structure of an agent program with an example agent program. (10 Marks)
b. With a neat diagram and agent function, explain the following :
i) Simple Reflex Agents ii) Goal Based Agents (10 Marks)

Module-2

- 3 a. Illustrate with an example, the components of a Well-defined problem. (10 Marks)
b. What are Toy problems and Real-world problems? Explain the formulation of vacuum world problem with a neat diagram of its state space. (10 Marks)

OR

- 4 a. Explain Breadth-First search strategy with a pseudocode for BFS on a graph and simple binary tree. (10 Marks)
b. What is Heuristic search? Explain greedy best-first search with an example. (10 Marks)

Module-3

- 5 a. Discuss different types of Machine Learning with example. (10 Marks)
b. Explain Machine Learning process with a neat diagram. (10 Marks)

OR

- 6 a. Write a note on :
i) Elements of big data ii) Types of data (10 Marks)
b. Explain 4-layer architecture of big data analysis framework. (04 Marks)
c. Write a note on Data Preprocessing step of Big Data processing cycle. (06 Marks)

Module-4

- 7 a. Explain Bivariate data with an example. (03 Marks)
b. Define covariance and correlation. Find the covariance and correlation coefficient of data.
 $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 4, 9, 16, 25\}$ (07 Marks)
c. Write the procedure for applying Gaussian elimination method. Solve the following set of equations using Gaussian elimination method:
$$\begin{matrix} 2x_1 + 4x_2 = 6 \\ 4x_1 + 3x_2 = 7 \end{matrix}$$
 (10 Marks)

OR

- 8 a. Write Find-S algorithm. Apply Find-S algorithm on the below dataset that contains details of the performance of students and their likelihood of getting a job offer or not in their final semester.

CGPA	Interactiveness	Practical Knowledge	Communication Skills
≥ 9	Yes	Excellent	Good
≥ 9	Yes	Good	Good
≥ 8	No	Good	Good
≥ 9	Yes	Good	Good
Logical Thinking	Interest	Job Offer	
Fast	Yes	Yes	
Fast	Yes	Yes	
Fast	No	No	
Slow	No	Yes	

(10 Marks)

- b. List the limitations of Find-S algorithm.

(03 Marks)

- c. Write K-Nearest-Neighbors (K-NN) algorithm.

(07 Marks)

Module-5

- 9 a. What are artificial neurons? Describe the structure of a single neuron with a neat diagram.

(04 Marks)

- b. Explain simple model of an artificial neuron.

(08 Marks)

- c. Write and explain Perceptron algorithm.

(08 Marks)

OR

- 10 a. Elaborate on the types of Artificial Neural Networks.

(10 Marks)

- b. Write and explain the algorithm for Radial Basis Function Neural Network.

(10 Marks)

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CBCS SCHEME

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21IS71

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Cryptography and Network Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Draw the simplified model of symmetric encryption and explain it. (06 Marks)
- b. Explain caesar cipher with example. (04 Marks)
- c. Explain playfair cipher algorithm. Find the cipher text for plain text = "instruments" with key = "MONARCHY". (10 Marks)

OR

- 2 a. Encrypt the plaintext "Cryptography" using Hill Cipher algorithm with key $K = \begin{bmatrix} 9 & 4 \\ 5 & 7 \end{bmatrix}$ and decrypt the same. (10 Marks)
- b. With a neat schematic diagram, explain the DES encryption algorithm. (10 Marks)

Module-2

- 3 a. With a neat diagram, explain the six ingredients of a public-key cryptography. (06 Marks)
- b. Explain the requirements and applications for public key cryptography. (04 Marks)
- c. Explain the Elgamal crypto system. (10 Marks)

OR

- 4 a. Explain RSA Algorithm. Using RSA algorithm perform encryption and decryption using $p = 17, q = 11, e = 7$ and $M = 88$. (10 Marks)
- b. Explain the Diffie-Hellman key exchange algorithm. (10 Marks)

Module-3

- 5 a. With a neat diagram, explain public key Authority and Public key certificates techniques for distribution of public keys. (10 Marks)
- b. Explain the key distribution scenario with relevant diagram. (10 Marks)

OR

- 6 a. Explain secret key distribution with confidentiality and authentication, with a neat diagram. (10 Marks)
- b. With a neat diagram, explain control vector Encryption and Decryption. (10 Marks)

Module-4

- 7 a. Describe Public key infrastructure, with neat diagram. (10 Marks)
- b. Explain Remote user – Authentication principles. (10 Marks)

OR

- 8 a. With a neat diagram, explain the general format of X.509 certificate. (10 Marks)
- b. Explain the differences between Kerberos version 4 and version 5 and also mention the technical deficiencies in Kerberos version 4 protocols. (10 Marks)

Module-5

- 9 a. Describe in detail PGP (Pretty Good Privacy) cryptographic functions. (10 Marks)
b. Describe the various header fields defined in MIME. (05 Marks)
c. List the important features of IKE key determination algorithm. (05 Marks)

OR

- 10 a. Explain the Applications and Benefits of IPsec. (10 Marks)
b. With a neat diagram, describe IKE header and payload format. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Advanced Mathematical Methods

Time: 3 hrs.

Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. Provide Normal distribution, t – distribution and F – distribution tables.

Module-1

- 1 a. Transform the matrix

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$$

into tri-diagonal form by Given's method. Using Strum's sequence,

obtain exact eigen values or the interval of unit length each containing one eigen value.

(10 Marks)

- b. Perform two iterations of the Bridge – Vieta method to find a real root of the equation $x^4 - 3x^3 + 3x^2 - 3x + 2 = 0$. Use the initial approximation $P_0 = +0.5$ and also find the deflated polynomial. (10 Marks)

OR

- 2 a. Transform the matrix

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 3 & 2 & -1 \\ 4 & 1 & 1 \end{bmatrix}$$

into tri – diagonal form by Householder's method. Using Strum's

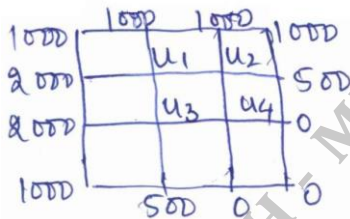
sequence, obtain exact eigen value or the interval of unit length each containing one eigen value.

(10 Marks)

- b. Perform two iterations of the Bairstow method to extract a quadratic factor $x^2 + px + q$ from the polynomial $x^3 + x^2 - x + 2 = 0$. Use the initial approximations $P_0 = -0.9$, $q_0 = 0.9$. (10 Marks)

Module-2

- 3 a. Solve the Laplace equation $U_{xx} + U_{yy} = 0$ for the square mesh of the below figure with boundary values as given below :



Carryout three iterations.

(10 Marks)

- b. Solve the Poisson's equation $U_{xx} + U_{yy} = -81xy$. Where $0 < x < 3$, $0 < y < 3$ given that $u(0, y) = 0$, $u(x, 0) = 0$, $u(3, y) = 100$, $u(x, 3) = 100$ and $h = 1$. (10 Marks)

OR

- 4 a. Use the Crank – Nicolson scheme to solve the heat equation problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \text{ with the conditions : } u(x, 0) = \sin x, \quad 0 \leq x \leq \pi, \quad u(0, t) = u(\pi, t), \quad t > 0.$$

(10 Marks)

- b. Obtain the solution of $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ by taking $h = 0.2$, where $0 \leq x \leq 1$, subjected to the initial conditions $u(x, 0) = \sin \pi x$, $u_t(x, 0) = \sin \pi x$ and the boundary conditions $u(0, t) = 0$, $u(1, t) = 0$ by using explicit method.

(10 Marks)

Module-3

- 5 a. Find the Taylor's expansion of $f(z) = \frac{2z^3 + 1}{z^2 + 1}$ in the region $z = i$. (06 Marks)
- b. Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $|z| = 3$. (07 Marks)
- c. Show that $\int_0^{2\pi} \frac{d\theta}{(1 - 2a \cos \theta + a^2)} = \frac{2\pi a^2}{(1 - a^2)^2}$, $a < 1$ by using contour integration. (07 Marks)

OR

- 6 a. Find the Laurent's expansion of $f(z) = \frac{7z - 2}{(z-1)(z-2)}$ about the point $1 < z < 2$. (06 Marks)
- b. Evaluate $\oint_C \frac{z-3}{(z^2 + 2z + 5)} dz$, where C is the circle $|z + 1 - i| = 2$. (07 Marks)
- c. Evaluate $\int_0^\infty \frac{\cos ax}{x^2 + 1} dx$ by using Contour integration. (07 Marks)

Module-4

- 7 a. Two independent sample of eight and seven items respectively had the following values of the variable :

Sample 1 :	9	11	13	11	15	9	12	14
Sample 2 :	10	12	10	14	9	8	10	

Do the two estimates of population variance differ significantly at 5% level of significance ($F_{0.05} = 4.21$). (10 Marks)

- b. Set up an analysis of variance (ANOVA) table to assess the significance of possible variation in performance in a certain test between the schools of a city. A common test was given to a numbers of students taken at random from the fifth class of the three schools concerned the results given below :

A	B	C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Construct the analysis of variance for the given data.

(10 Marks)

OR

- 8 a. The mean and S.D of the maximum loads supported by 60 cables are 11.09 tonnes and 0.73 tonnes respectively. Find i) 95% ii) 99% confidence limits for mean of the maximum loads of all cables produced by the company. (10 Marks)
- b. Measurements on the length of a copper were taken by two experiments A & B are as under

A's measurements	12.29	12.25	11.86	12.13	12.44	12.78	12.77	11.90	12.47
B's measurements	12.39	12.46	12.34	12.22	11.98	12.46	12.23	12.06	

Test whether B's measurements are more accurate than A's at 5% level of significance.

(10 Marks)

Module-5

- 9 a. Explain i) Absorbing state ii) Transient state iii) Recurrence state. (06 Marks)
- b. Three boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. But C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball, find the probability that A has the ball for the forth throw. (07 Marks)
- c. Customers arrive at a one – man barber shop according to a Poisson process with a mean inter arrival time of 12 min. Customers spend an average of 10mm in the barber's chair
- i) What is the expected number of customers in the barber shop and in queue?
- ii) Calculate the percentage of time an arrival can walk straight into barber's chair without having to wait. (07 Marks)

OR

- 10 a. Find the unique fixed probability vector for the regular stochastic matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

(06 Marks)

- b. A student's study habits are as follows. If he studies one night , he is 70% sure not to study the next night. On the other hand , if he does not study one night , he is 60% sure not to study the next night. In the long run how often does he study? (07 Marks)
- c. Given an average arrival rate of 20 per hour, is it better for a customer to get service at a single channel with mean service rate of 22 customers per hour or at one of two channels is parallel with mean service rat at 11 customers per hour for each of the two channels. Assume both the queues to be Poisson type. (07 Marks)

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Research Methodology and Intellectual Property Rights

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Give various definitions of research explaining its meaning and scope. Explain the objective and motivation in Engineering research. (08 Marks)
- b. Describe the steps involved in research process with neat diagram. (06 Marks)
- c. What is ethics in Engineering research? Why it is important. (06 Marks)

OR

- 2 a. What is research problem? Define the main issues which should receive the attention of the research in formulating the research problem. (08 Marks)
- b. Discuss in detail various types of research. (06 Marks)
- c. Describe different types of research misconduct. (06 Marks)

Module-2

- 3 a. Discuss the importance of critical literature review and its uses in planning innovation research. (06 Marks)
- b. Explain the concept of knowledge flow through citation. How collaborations certainly impact citation counts? Explain. (08 Marks)
- c. Write short notes on : (06 Marks)
 - i) Google and Google scholar
 - ii) Acknowledgment and Attributions

OR

- 4 a. Analyze the following terms which do not fulfill the actual goal of citations and acknowledgements. i) Spurious citations ii) Biased citations iii) Self citations iv) Coercive citations. (08 Marks)
- b. What are the things author should acknowledge? (06 Marks)
- c. Elaborate on the following : (06 Marks)
 - i) Technical reading
 - ii) Critical and creative reading.

Module-3

- 5 a. What are Intellectual Property Rights? Explain the necessity of it. (06 Marks)
- b. Differentiate between invention and Innovation. What is a Patent? What are the criteria of patentability? (08 Marks)
- c. Enumerate the procedure for application preparation filing and grant of patents. (06 Marks)

OR

- 6 a. List and explain in detail about various types of Intellectual property rights. (08 Marks)
- b. Explain in detail about the infringement of patents. (06 Marks)
- c. What types of invention are not patentable in India? (06 Marks)

Module-4

- 7 a. What is Copyright? Explain classes, criteria and ownership of copyright. (08 Marks)
 b. Discuss about the copyright ownership issues. (06 Marks)
 c. Describe the process involved in the registration of a trademark. (06 Marks)

OR

- 8 a. Enumerate the procedure for registration of copyright. (08 Marks)
 b. Explain about the Trademark and rights from trademarks registration. (06 Marks)
 c. Explain about types of Trademarks registered in India. (06 Marks)

Module-5

- 9 a. How can industrial designs be protected? (06 Marks)
 b. What is a geographical indication? Explain the following with respect to GI.
 i) Ownership and right granted to the GI holders
 ii) Registered GI in India (08 Marks)
 c. Explain the following :
 i) Famous Industrial Designs
 ii) Generic GI and Homonymous GI (06 Marks)

OR

- 10 a. Explain the classification of Industrial Designs. (06 Marks)
 b. Explain the procedure for GI registration. (08 Marks)
 c. Explain about the enforcement of IPR in India. (06 Marks)

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18CS32

Third Semester B.E. Degree Examination, Dec.2024/Jan.2025

Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Illustrate the difference between linear and non – linear data structure with an example. (04 Marks)
b. Write the functions to compare and concatenate strings. What are the equivalent library functions? (08 Marks)
c. Illustrate with a diagram 2 – d array of 3 – rows and 4 – columns in row major order. Write a function to print row and column index of a given number N. (08 Marks)

OR

- 2 a. Write the type definition for an element as triplets in a sparse matrix. What is the advantage? (04 Marks)
b. With a diagram, illustrate representation of the polynomial $A(x) = a_0 + a_1x^1 + a_2x^2 + \dots + a_nx^n$. Write a program to construct the polynomial. (08 Marks)
c. Implement C - program to match all occurrence of pattern 'P' in text 'T'. (08 Marks)

Module-2

- 3 a. Write a recursive function to compute gcd of two numbers. (04 Marks)
b. Write a program to implement the stack operations using dynamic array. (08 Marks)
c. Convert $a * x^2 + b * x + C$ into post fix expression using stack. (08 Marks)

OR

- 4 a. Write a recursive function for Ackermann function $A(m, n)$.
$$A(m, n) = \begin{cases} n+1 & \text{if } m = 0 \\ A(m-1, 1) & \text{if } m \neq 0 \text{ but } n = 0 \\ A(m-1, A(m, n-1)) & \text{if } m \neq 0 \text{ \& } n \neq 0 \end{cases}$$
 (04 Marks)
b. Write a program to implement queue operations for the circular queue. (08 Marks)
c. Evaluate postfix expression $ab + cd + * 2 /$, when $a = b = c$ and $c = d = 4$ using stack. (08 Marks)

Module-3

- 5 a. Describe the dynamic memory allocation and free functions. (04 Marks)
b. Write a program to create linked list for the polynomial $1 + 2x + 3x^2 + \dots + nx^{n-1}$. (08 Marks)
c. Write a program to implement stack operations using linked list. (08 Marks)

OR

- 6 a. Define self referential structure for integer data and allocate the memory required by the structure. (04 Marks)
- b. Given a linked list of characters (data field) 'A', 'B' and 'C'. Write a function to
 i) insert 'X' between 'A' and 'B' ii) Delete 'B' from the list. (08 Marks)
- c. Write a program to implement queue operations using linked list. (08 Marks)

Module-4

- 7 a. What is Binary Search Tree (BST)? Define structure of a node in BST in 'C'. (04 Marks)
- b. Draw an expression tree for $((a + (b * ((c - d) / e))) + f)$. (08 Marks)
- c. Write a function to insert an item into a binary search tree without duplications. (08 Marks)

OR

- 8 a. Explain strictly binary tree and complete binary tree with examples for each. (04 Marks)
- b. Given pre – order sequence 1, 2, 4, 3, 5, 7, 8, 6 and inorder sequence 4, 2, 1, 7, 5, 8, 3, 6. Construct a corresponding BST. (08 Marks)
- c. Illustrate depth of the tree with an example. Write a function to find the depth of the tree. (08 Marks)

Module-5

- 9 a. With an example, describe two representations of the graph. (04 Marks)
- b. Explain graph traversal methods. Write a program for depth first search. (08 Marks)
- c. Describe the types, attributes and operations. (08 Marks)

OR

- 10 a. Describe hash table including collision and overflows. (04 Marks)
- b. Write an algorithm for insertion sort. Apply insertion sort showing various passes to sort the array $A = [2, 1, 6, 5, 4, 3]$. (08 Marks)
- c. What is a file? Describe sequential and index sequential file organizations. (08 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025

Analog and Digital Electronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the construction, working and V-I characteristics of photo diode. (06 Marks)
- b. List the different types of BJT biasing. Derive the expression for fixed bias circuit. (06 Marks)
- c. With the help of neat circuit diagram and waveform explain the working principle of relaxation oscillator. (08 Marks)

OR

- 2 a. With a neat waveform and circuit diagram, explain the working of monostable multivibrator. (06 Marks)
- b. Explain the working of R-2R ladder D to A converter. (06 Marks)
- c. Explain performance parameters of a power supply. (08 Marks)

Module-2

- 3 a. A logic circuit realizing the function f has four inputs A, B, C and D. The three inputs A, B and C are the binary representation of the digits 0 through 7 with A being the most significant bit. The input D is an odd parity bit, i.e., the value of D is such that A, B, C and D always contain an odd number of 1's. (For example, the digit 1 is represented by ABC = 001 and D = 0, and the digit 3 is represented by ABCD = 0111). The function f has value 1 if the input digit is an odd number and value 0 if the input digit is an even number.
 - i) List the minterms and don't-care minterms of f in algebraic form.
 - ii) List the maxterms and don't-care maxterms of f in algebraic form. (10 Marks)
- b. Solve for the Boolean expression using K-map to get minimum SOP and minimum POS expression:
 - i) $f(A, B, C, D) = \sum m(0, 1, 4, 8, 9, 10) + d(2, 11)$
 - ii) $f(A, B, C, D) = \pi M(0, 1, 3, 4, 7)$ (10 Marks)

OR

- 4 a. Determine the essential prime implicants using Q method for the function $f(A, B, C, D) = \sum m(0, 1, 2, 3, 10, 12, 11, 13, 14, 15)$. (10 Marks)
- b. Explain with explain Petrik's method. (06 Marks)
- c. Find the minimum SOP for the function $f(A, B, C) = \sum m(2, 6, 7)$ using Quine McCluskey method. (04 Marks)

Module-3

- 5 a. How do you detect static hazards in combinational logic circuits? Explain with examples. (06 Marks)
- b. With a neat diagram, explain 3 to 8 line decoder. (06 Marks)
- c. Design 7 segment decoder and realize using PLA. (08 Marks)

OR

- 6 a. Implement $Y(A, B, C, D) = \sum m(0, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13, 15)$ using 16 – to – 1 multiplexer (IC 74150) and 8 – to – 1 multiplexer. (08 Marks)
- b. Implement full adder using PAL. (08 Marks)
- c. Explain simulation and testing of digital circuits. (04 Marks)

Module-4

- 7 a. Explain the structure of VHDL program. Write VHDL code for 4-bit parallel adder using full adder as component. (08 Marks)
- b. Construct SR latch using NAND gates and derive the characteristics equation for the same. (08 Marks)
- c. Differentiate between latch and flip flop. Show how SR flip flop can be converted to D flip flop. (04 Marks)

OR

- 8 a. Derive the characteristics equations for D, T, SR and JK flipflops. (08 Marks)
- b. Explain the working of a JK flip flop. Write its Truth table, state diagram and excitation table. (08 Marks)
- c. Give the implementation of T flip flop from D flip flop. (04 Marks)

Module-5

- 9 a. What is register? Explain how 4 bit register with data, load, clear and clock input is constructed using D flip-flops. (06 Marks)
- b. With a neat diagram, explain 4 bit SISO register. (06 Marks)
- c. Design a random counter using T flip flops whose transition graph is shown in Fig.Q.9(c).

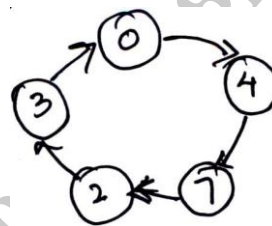


Fig.Q.9(c)

(08 Marks)

OR

- 10 a. With the help of state graph, state and transistor tables and timing diagrams explain sequential parity checker. (08 Marks)
- b. Differentiate between Moore and Melay machines. Analyze following Moore sequential circuit for an input sequence of $X = 01101$ and draw the timing diagram. (08 Marks)
- c. Write a note on Ring counter. (04 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Computer Organization

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat diagram, explain the basic operational concepts of a computer. (06 Marks)
- b. Explain the basic instruction types with example. (08 Marks)
- c. Explain Big Endian and Little Endian methods by addressing with proper example. (06 Marks)

OR

- 2 a. Define addressing mode. Explain the following addressing modes with suitable examples :
 - (i) Register
 - (ii) Direct
 - (iii) Indirect
 - (iv) Auto-Increment
- b. Explain SPEC rating of a computer. (10 Marks)
- c. Explain SPECT rating of a computer. (05 Marks)
- c. Write an assembly program to add numbers stored in consecutive memory locations as Num₁, Num₂,..... Num_n and store the result at location SUM by using branching technique. (05 Marks)

Module-2

- 3 a. What is DMA? Explain its operation using registers in a DMA interface. (08 Marks)
- b. Explain the following interrupt nesting methods :
 - (i) Daisy chain.
 - (ii) Implementation of interrupt nesting.
- c. What is DMA Bus Arbitration? Explain Distributed arbitration with neat diagram. (06 Marks)

OR

- 4 a. Explain the I/O interface for an input device (key board) interface to the processor with a neat block diagram. (08 Marks)
- b. Explain synchronous Bus with neat timing diagram. (06 Marks)
- c. With neat diagram, explain USB split bus operation. (06 Marks)

Module-3

- 5 a. With a neat diagram, explain the internal organization of 2M × 8 DRAM chip. (10 Marks)
- b. Explain Direct Mapping function. (05 Marks)
- c. Explain the read and write operation in a single SRAM cell with circuit diagram. (05 Marks)

OR

- 6 a. With neat diagram, explain the memory hierarchy with respect to speed, size and cost. (05 Marks)
- b. With neat circuit diagram, explain how data are written into ROM. (05 Marks)
- c. What is memory interleaving? Explain with a suitable example. (10 Marks)

Module-4

- 7 a. Explain the design of a 4-bit carry look ahead adder. Calculate number of gate delays for S_3 and C_4 . (08 Marks)
- b. With neat diagram, explain binary addition/Subtraction. (06 Marks)
- c. Multiply (+14) and (-6) using Booth's algorithm. (06 Marks)

OR

- 8 a. Perform bit pair recoding for -11 and +27 [(-11) multiplicand and (+27) multiplier]. (06 Marks)
- b. Perform the following operations on the 5-bit signed numbers using 2's complement representation system. Further indicate whether overflow has occurred :
- (i) $(-10) + (-13)$
- (ii) $(+10) - (+14)$
- (iii) $(+7) - (-15)$ (06 Marks)
- c. Perform $1100 \div 11$ using non restoring division algorithm. (08 Marks)

Module-5

- 9 a. With neat diagram, explain the working of single-bus organization of the datapath inside processor. (08 Marks)
- b. Write the sequence of control steps for the execution of an instruction Add (R_3), R_1 . (07 Marks)
- c. Write the control sequence for the execution of an unconditional branch instruction. (05 Marks)

OR

- 10 a. With neat diagram, explain the organization of Hardwired control unit. (07 Marks)
- b. What is pipelining? Explain the 4-stage pipeline. (05 Marks)
- c. With neat diagram, explain the Three-bus organization of the datapath. (08 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Software Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Software Engineering? Explain software engineering code of Ethics. (10 Marks)
- b. With a neat diagram, explain the waterfall model of software development process. (10 Marks)

OR

- 2 a. With a neat block diagram, explain the spiral processing model. (10 Marks)
- b. What are attributes of good software? Explain the key challenges facing software engineering. (10 Marks)

Module-2

- 3 a. Explain in brief links, association, ordering bags and sequences with an example each. (10 Marks)
- b. Explain following important terms with example : (10 Marks)
 - i) Identify
 - ii) Classification
 - iii) Inheritance
 - iv) Polymorphism.

OR

- 4 a. Explain generalization and inheritance with an example each. (10 Marks)
- b. Explain in brief class model, state model, and interaction model with an example each. (10 Marks)

Module-3

- 5 a. What is system modelling? Explain the different perspective that the system model developed. (10 Marks)
- b. Explain model driven engineering in details and mention advantages and disadvantages. (10 Marks)

OR

- 6 a. With a neat diagram, explain context model with an example. (10 Marks)
- b. Explain event-driven model with a state diagram of microwave over application. (10 Marks)

Module-4

- 7 a. What is test driven development? With a neat diagram, explain test driven development process. (10 Marks)
b. With appropriate block diagram, explain the system evolution process. (10 Marks)

OR

- 8 a. Write and explain Lehman's laws related to system changes. (10 Marks)
b. Explain reengineering process with a neat block diagram. (10 Marks)

Module-5

- 9 a. With a neat diagram, explain project scheduling process. (10 Marks)
b. List and explain factors affecting software pricing. (10 Marks)

OR

- 10 a. Discuss in detail the different stages in component measurement process with diagram. (10 Marks)
b. Explain plan driven development with a neat block diagram. (10 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025
Computer Networks and Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With neat diagram, explain HTTP Request and Response messages. (10 Marks)
 b. Illustrate the basic operations of SMTP with diagram. (10 Marks)

OR

- 2 a. Explain FTP with its commands and replies. (10 Marks)
 b. With a diagram, explain the interaction of various DNS servers. (10 Marks)

Module-2

- 3 a. Describe the various fields of UDP segment. Explain how checksum is calculated. (10 Marks)
 b. With a diagram, explain the FSM for sender and receiver side of rdt 2.0 protocol. (10 Marks)

OR

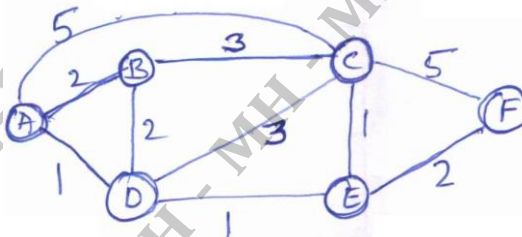
- 4 a. Explain 2 way handshake and closing a TCP connection with diagram. (10 Marks)
 b. With neat diagram, explain TCP segment structure and its field. (10 Marks)

Module-3

- 5 a. With diagram, explain router architecture. (06 Marks)
 b. With general format, explain various fields of IPV6 datagram. (08 Marks)
 c. Compare IPV4 and IPV6 datagram. (06 Marks)

OR

- 6 a. Write the link state algorithm and apply it to the following graph with source node A. (10 Marks)



- b. Briefly illustrate the Routing Information Protocol (RIP). (06 Marks)
 c. Briefly explain the Border Gateway Protocol (BGP). (04 Marks)

Module-4

- 7 a. What are the elements of Network security? Explain the threats to Network security. (08 Marks)
 b. Discuss about : i) Cryptographic technique ii) Authentication technique. (06 Marks)
 c. Briefly explain the steps of DES algorithm. (06 Marks)

OR

- 8 a. Explain RSA algorithm using RSA algorithm encrypt a message $m = 9$. Assume $P = 5$ and $q = 11$. Find the public and private keys and also show the cipher text. (10 Marks)
- b. Discuss the following : i) Secure Hash Algorithm ii) Firewalls. (10 Marks)

Module-5

- 9 a. Briefly explain the properties of Audio and Video. (08 Marks)
- b. List the categories of streaming of stored video and explain any one. (08 Marks)
- c. Classify the multimedia network applications. (04 Marks)

OR

- 10 a. With neat diagram, explain CDN operations. (10 Marks)
- b. With general format, explain the various fields of RTP header. (10 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025
Database Management Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the main characteristics of the database approach. How does it differ from traditional file systems? (08 Marks)
- b. With neat diagram, explain three schema architecture. (06 Marks)
- c. What are the different types of database end users? Discuss the main activities of each. (06 Marks)

OR

- 2 a. Design an ER diagram for company database with atleast four entities. (08 Marks)
- b. Define the following with an example :
 - i) Weak entity type ii) Participation constraints iii) Cardinality ratio
 - iv) Recursive relationship. (08 Marks)
- c. What is Generalization? Illustrate how it is helpful with an example. (04 Marks)

Module-2

- 3 a. What is meant by integrity constraint? Explain the importance of referential integrity constraint. How referential integrity constraint is implemented in SQL? (08 Marks)
- b. Explain with examples in SQL :
 - i) Drop command ii) Delete command iii) Update command. (06 Marks)
- c. Discuss various types of inner join operations. (06 Marks)

OR

- 4 a. Write the relational algebra operations to perform the following queries :
 - i) Retrieve the name and address of all employees who work for the "Accounts" department.
 - ii) Retrieve the names of employees who have no dependents.
 - iii) Find the names of employees who work on all the projects controlled by department number 2. (12 Marks)
- b. Describe the steps of an algorithm for ER – to – Relational mapping. (08 Marks)

Module-3

- 5 a. How are triggers and assertions defined in SQL? Explain. (08 Marks)
- b. How are views created and dropped? Explain how the views are implemented and updated. (08 Marks)
- c. What is SQLJ? How it is different from JDBC? (04 Marks)

OR

- 6 a. With an example, explain stored procedures in SQL. (06 Marks)
b. Draw and explain three – tier architecture and technology relevant to each tier. Write the advantages of three – tier architecture. (06 Marks)
c. What are the components of the JDBC architecture? Describe four different architecture alternations for JDBC drivers. (08 Marks)

Module-4

- 7 a. Define Normal form. Explain 1NF , 2NF and 3NF with suitable example. (08 Marks)
b. Write an algorithm to find the closure of 'X' and 'F'. (06 Marks)
c. Explain any two informal quality measures employed for a relational schema design. (06 Marks)

OR

- 8 a. Explain the 4NF with a suitable example. (08 Marks)
b. Write an algorithm for testing non additive join property. (08 Marks)
c. Explain Armstrong inference rules. (04 Marks)

Module-5

- 9 a. Why concurrency control is needed? Demonstrate with an example. (10 Marks)
b. Discuss the ACID properties of a database transaction. (04 Marks)
c. Discuss the UNDO and REDO operations and the recovery techniques that use each. (06 Marks)

OR

- 10 a. Discuss Two – Phase locking technique for concurrency control. (10 Marks)
b. Explain transaction supports in SQL. (06 Marks)
c. Explain how shadow paging helps to recover from transaction failure. (04 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Automata Theory and Computability

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following with example: i) Alphabet ii) Power of an alphabet iii) String and length of string iv) Concatination v) Language. (10 Marks)
- b. Define DFSM. Design a DFSM to accept each of the following languages:
 $L = \{W : |W| \bmod 3 = 0\}$ where $\Sigma = \{a\}$
 $L = \{W \in \{a, b\}^* : W \text{ Ending with } abb\}$ (10 Marks)

OR

- 2 a. Define NDFSM. Design an NFA to recognize the following set of string : [0101, 101, 011]. (06 Marks)
- b. Design an NFA to obtain string of a and b ending with ab (or) ba, convert it into its equivalent DFA. (08 Marks)
- c. Differentiate between DFA, NFA and ϵ NFA. (06 Marks)

Module-2

- 3 a. Define regular expression, write the regular expression for the following languages:
 i) $\{W \in \{0,1\}^* \mid |W| \text{ is even}\}$
 ii) $\{W \in \{0,1\}^* \mid \text{has } 001 \text{ as substring}\}$
 iii) $\{W \in \{a,b\}^* \mid \text{whose second symbol from right end is } a\}$
 iv) $\{W \in \{a,b\}^* \mid \text{starting with 'a' and ending with } b\}$ (06 Marks)
- b. Show that every regular expression there is an equivalent FSM. (06 Marks)
- c. Construct FSM for the regular expression
 i) $a^* + b^* + c^*$ ii) $(a + b)^* aa (a + b)^*$ (08 Marks)

OR

- 4 a. State and prove pumping lemma theorem for RL and ST the language $L = \{a^i b^j : i, j \geq 0 \text{ and } i - j = 5\}$ is not regular. (12 Marks)
- b. List the closure properties of regular language. Explain any two of them with example. (08 Marks)

Module-3

- 5 a. Define context free grammer. Design CFG for the following languages:
 i) Let $\Sigma = \{a, b\}$ to generate string of even number of a's.
 ii) $L = \{a^n \cdot b^n \mid n \geq 0\}$
 iii) To generate string consisting of multiples of three a's. (10 Marks)
- b. Obtain the grammer to generate the following language:
 i) $L = \{0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0\}$
 ii) $L = \{a^i b^j \mid i \neq j, i \geq 0 \text{ and } j \geq 0\}$ (10 Marks)

OR

- 6 a. What is ambiguity? Show that the following grammar is ambiguous
 $S \rightarrow AB \mid aaB$
 $A \rightarrow a \mid Aa$
 $B \rightarrow b$
 (06 Marks)
- b. Define push down automata. Obtain a PDA to accept the language $L = A^n B^n = \{a^n b^n : n \geq 1\}$
 (08 Marks)
- c. i) Derive leftmost derivation for the string aaabbabbba using the following grammar
 $S \rightarrow aB \mid bA$
 $A \rightarrow aS \mid bAA \mid a$
 $B \rightarrow bS \mid aBB \mid b$
 ii) Obtain the rightmost derivation for the string $id + id * id$ using
 $E \rightarrow E + E$
 $E \rightarrow E * E$
 $E \rightarrow E - E$
 $E \rightarrow E/E$
 $E \rightarrow id$
 (06 Marks)

Module-4

- 7 a. Define Turing machine model. Explain representation of Turing machine. (08 Marks)
 b. Design a Turing machine to accept $L = \{0^n 1^n 2^n \mid n \geq 1\}$ (08 Marks)
 c. Write a short note on multi tape TM. (04 Marks)

OR

- 8 a. With neat diagram, explain variants of TM. (10 Marks)
 b. Explain the model of linear bound automation. (05 Marks)
 c. Explain the working of a Turing machine. (05 Marks)

Module-5

- 9 a. Explain the following with example:
 i) Decidability
 ii) Decidable language
 iii) Undecidable language. (06 Marks)
 b. Explain post correspondence problem. (07 Marks)
 c. Explain halting problem in TM. (07 Marks)

OR

- 10 Write a short note on:
 a. Growth rate of function
 b. Classes of P and NP
 c. Quantum computers
 d. Church-Turing thesis. (20 Marks)

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Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Application Development using Python

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What do all expressions do? Explain the math operators in python from highest to lowest precedence with an example for each. Write steps involved in python for evaluation of the expression, $(s-1)*\left(\frac{(7+1)}{(3-1)}\right)$ and reduce it to a single value. (06 Marks)
- b. Explain python supported flow control statements with an example program. (08 Marks)
- c. Write a python program to extract last digit of a given number. (06 Marks)

OR

- 2 a. Explain parameter passing of user-defined function in python with an example program. (09 Marks)
- b. How many global scopes are there in a python program? How many local scopes? What happens to variables in a local scope and global scope when the function call returns, illustrate with a program consisting of minimum one local and global variable. (05 Marks)
- c. What is exception handling? How exceptions are handled in python? Write a Python program with exception handling code to solve divide-by-zero error situation. (06 Marks)

Module-2

- 3 a. List any four python string method with a suitable example. (10 Marks)
- b. Compare list and string. (06 Marks)
- c. Write a python program to determinate tuple by sorting given words from longest to shortest using loops and also remove all duplicate letters in a given word. (04 Marks)

OR

- 4 a. Describe converting types with list and tuple with an example. (06 Marks)
- b. Mention merits of list over dictionary. Write a python program to accept USN and marks obtained. Find maximum, minimum and students USN who have scored in the range 100-85, 85-75, 75-60 and below 60. (08 Marks)
- c. Explain the usage of 'in' and 'not in' operators with string in python through an example. (06 Marks)

Module-3

- 5 a. Describe python instruction for the following in reference to logging.
debug(), (i) Logging levels (ii) Send login message to file named exam.net.
(iii) No debug with print (09 Marks)
- b. Explain fileopen, fileclose, fileread and filewrite functions of python with an example program. (05 Marks)
- c. Write a python program to search for word start with 'The' followed by two words and ends with 'es' in a given string. (06 Marks)

OR

- 6 a. Describe OS.path modules. Also discuss the following methods of OS module :
 (i) chdir()
 (ii) unlink
 (iii) walk
 (iv) listdir() (10 Marks)
- b. Differentiate between shutil.copy() and shutil.copytree() with a python code. (06 Marks)
- c. Write a python program to search for a word that starts with 'A' followed by three or four letters followed by 'u'. (04 Marks)

Module-4

- 7 a. Demonstrate polymorphism with function to find histogram to count the number of times a letter appears in word and sentence. Create a class student with following member attributes : roll no, name, age and total marks. Create suitable methods for reading and printing member variables. Write a python definition in method to overload '=' operator to print the details of student having same marks. (10 Marks)
- b. Define pure function with an example. Write a program to express instance as return values to define a class CIRCLE with members radius, origin-X, origin-Y and member functions : to find area of circle, perimeter of circle and to relocate circle by 10 points vertically and 5 points horizontally. (10 Marks)

OR

- 8 a. Define method and positional argument with reference to class. Create a class : Employee to store personnel details. Such as id, name, gender, city and salary and define methods to initialize and display employee details. (10 Marks)
- b. Describe on programmer-defined type. Write an object oriented python program to create two objects of Time : Current_time and Bread_time which contains bread baking time. Include addTime method to display the total time taken by the bread maker to prepare bread. (10 Marks)

Module-5

- 9 a. Explain requirement of openpyxl. In a worksheet object, write python code to retrieve workbook active sheet and to set value in the cell C₅ to "5th semester", also to update spreadsheet of marks using updated marksheet.xsl file. (10 Marks)
- b. A string value of the PDF filename is not passed to the PyPDF2.pdfFileReader () function. What is required to pass to the function instead? Describe python support for a word document to read and write from word document. (10 Marks)

OR

- 10 a. List a few features that openpyXL 2.1.4. Use functions and methods of OpenpyXL to create a bar chart for data stored in spread sheet to develop a python program. (10 Marks)
- b. How to view the HTML of a specific element on a web page? Mention steps to download all XKCD comics with a suitable python instruction. (10 Marks)

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Seventh Semester B.E. Degree Examination, Dec.2024/Jan.2025
User Interface Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain in detail the characteristics of Graphical user Interface (any 5). (10 Marks)
 b. Explain the advantages and disadvantages of Graphical System. (10 Marks)

OR

- 2 a. Explain 10 - differences between Graphical user Interface and Web page design. (10 Marks)
 b. Discuss the general principles of User Interface design (any 10). (10 Marks)

Module-2

- 3 a. List and explain five commandments in designing for people. (10 Marks)
 b. Describe the important human characteristics of UID. (10 Marks)

OR

- 4 a. Explain common usability problem in Web based system. (10 Marks)
 b. Explain techniques for determining user requirement using Indirect methods. (10 Marks)

Module-3

- 5 a. Explain in detail the structure of menu. (10 Marks)
 b. List all kinds of graphical menu and explain any two in detail. (10 Marks)

OR

- 6 a. Describe the functions of menu. (10 Marks)
 b. List components of web navigation system. (10 Marks)

Module-4

- 7 a. Explain the component of windows with example. (10 Marks)
 b. Discuss briefly about types of window with example. (10 Marks)

OR

- 8 a. Describe overlapping windows and tiled window presentation styles with example. (10 Marks)
 b. Explain characteristics of touch screen and keyboard. (10 Marks)

Module-5

- 9 a. Explain Radio – button and list boxes selection control. (10 Marks)
 b. Explain any 2 types of testing proto types used in UID. (10 Marks)

OR

- 10 a. Write a note on presentation control in UID. (10 Marks)
 b. Explain usability tests conducted in UID. (10 Marks)

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18CS745

Seventh Semester B.E. Degree Examination, Dec.2024/Jan.2025 Robotic Process Automation Design and Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain any eight benefits of RPA. (08 Marks)
- b. Illustrate the issues in Artificial Intelligence. (08 Marks)
- c. Explain any four downsides of RPA. (04 Marks)

OR

- 2 a. Illustrate the different cloud computing approaches and the associated drawbacks. (06 Marks)
- b. Explain Agile, Scrum, Kanban and Waterfall software management approaches. (08 Marks)
- c. Illustrate the different BPO strategies along with their drawbacks. (06 Marks)

Module-2

- 3 a. Explain components of RPA in detail with diagram. (10 Marks)
- b. Explain different RPA platforms in detail. (10 Marks)

OR

- 4 a. List and explain types of Robots. (06 Marks)
- b. Explain step by step following examples using recorders :
i) Emptying the trash folder in Gmail ii) Emptying Recycle Bin. (14 Marks)

Module-3

- 5 a. Define sequence. Illustrate the use of Arguments, variables and scope in a sequence. (08 Marks)
- b. Explain with neat diagram, different types of control flow activities. (08 Marks)
- c. Explain Read and Write cell with example. (04 Marks)

OR

- 6 a. Explain flowchart and usage of activities with workflows. (08 Marks)
- b. Create a database with the columns [Name, USN, Sem, mark1, marks2, mark3 and Total]. Calculate the total marks for 10 student and display the result. (08 Marks)
- c. Explain Append range and Write Range with example. (04 Marks)

Module-4

- 7 a. Explain any four activities that can be used to find the controls on screen. (08 Marks)
- b. Explain the techniques through which we can wait for a control. (06 Marks)
- c. Explain the Mouse activities in UiPath studio. (06 Marks)

OR

- 8 a. Illustrate screen scraping, and explain the different screen scraping methods. (08 Marks)
- b. Explain the different types of events. (06 Marks)
- c. Explain the keyboard activities in UiPath Studio. (06 Marks)

Module-5

- 9 a. Explain Exception Handling in RPA with example. (10 Marks)
b. Explain Logging and taking screen shots in RPA, with example. (10 Marks)

OR

- 10 Write short notes on :
a. Debugging techniques
b. Collecting crash dumps.
c. Error Reporting
d. Future of RPA.

(20 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Transform Calculus, Fourier Series and Numerical Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1 a. Find $L\left\{\frac{\cos at - \cos bt}{t}\right\}$. (06 Marks)

b. A periodic function of period $\frac{2\pi}{\omega}$ is defined by, $f(t) = \begin{cases} E \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$ (07 Marks)

c. Solve : $y' - y' - 2y = 0$; given $y(0) = 0$ and $y'(0) = 6$ by Laplace transformation method. (07 Marks)

OR

2 a. Find $L^{-1}\left\{\frac{4s+5}{(s-1)^2(s+2)}\right\}$. (06 Marks)

b. Apply convolution theorem to evaluate $L^{-1}\left\{\frac{1}{(s^2+1)(s^2+9)}\right\}$. (07 Marks)

c. Using unit step function, find the Laplace transform of, $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ \sin 2t, & \pi \leq t \leq 2\pi \\ \sin 3t, & t \geq 2\pi \end{cases}$ (07 Marks)

Module-2

3 a. Obtain the Fourier Series for the function, $f(x) = x^2$, $0 < x < 2\pi$. (06 Marks)

b. Find the Fourier series of $f(x)$,
Where $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$. (07 Marks)

c. Express $f(x) = x$ as a half-range cosine series in $0 < x < 2$. (07 Marks)

OR

4 a. Obtain Fourier series for the function, $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2 \end{cases}$. (06 Marks)

b. Find the Fourier half-range cosine series of the function $f(x) = (x+1)$, in $(0, 1)$. (06 Marks)

c. Compute the first harmonic of the Fourier series of $f(x)$ given in the following table :

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

(08 Marks)

Module-3

- 5 a. Find the z-transform of, $3n - 4\sin\left(\frac{n\pi}{4}\right) + 5a$. (06 Marks)
- b. Compute the inverse z-transform of, $\frac{2z^2 + 3z}{(z+2)(z-4)}$. (07 Marks)
- c. Find the Fourier transform of, $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$, Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$. (07 Marks)

OR

- 6 a. Find the Fourier sine transform of e^{-ax} . (06 Marks)
- b. If $U(z) = \frac{2z^2 + 5z + 14}{(z-1)^4}$, evaluate u_2 and u_3 . (07 Marks)
- c. Using the z-transform, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$, $u_0 = 0$, $u_1 = 1$. (07 Marks)

Module-4

- 7 a. Find by Taylor's series method the value of y at $x = 0.1$ and $x = 0.2$ to four places of decimals from, $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$. (06 Marks)
- b. Apply Runge-Kutta fourth order method to find an approximate value of y when $x = 0.2$, given $\frac{dy}{dx} = x + y$, $y(0) = 1$. (07 Marks)
- c. If $\frac{dy}{dx} = 2e^x - y$, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.04$ and $y(0.3) = 2.09$, find $y(0.4)$ by employing the Milne's predictor-corrector formula, use corrector formula twice. (07 Marks)

OR

- 8 a. Using modified Euler's method, solve the IVP $\frac{dy}{dx} = x + \sqrt{y}$, $y(0) = 1$ at $x = 0.2$, perform three modifications. (06 Marks)
- b. Using the fourth order Runge-Kutta method, solve the IVP $\frac{dy}{dx} = \frac{1}{x+y}$ at the point $x = 0.5$. Given that $y(0.4) = 1$. (07 Marks)
- c. Given $\frac{dy}{dx} = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$. Determine $y(1.4)$ by Adams-Bashforth method. (07 Marks)

Module-5

- 9 a. Using Runge-Kutta method of fourth order solve the differential equation, $\frac{d^2y}{dx^2} = x \left(\frac{dy}{dx} \right) - y^2$, with $y(0) = 1$, $y'(0) = 0$ at $x = 0.2$. (06 Marks)
- b. Derive Euler's equation in the standard form, $\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$. (07 Marks)

- c. On which curve the functional,

$$\int_0^{\frac{\pi}{2}} (y'^2 - y^2 + 2xy) dy, y(0) = 0, y\left(\frac{\pi}{2}\right) = 0 \text{ be extremized.}$$

(07 Marks)

OR

- 10 a. Apply Milne's method to compute $y(0.8)$, given that $\frac{d^2y}{dx^2} = 1 - 2y \frac{dy}{dx}$ and

x	0	0.2	0.4	0.6
y	0	0.02	0.0795	0.1762
y'	0	0.1996	0.3937	0.5689

(06 Marks)

- b. Prove that the geodesics on a plane are straight line.

(07 Marks)

- c. Find the extremal of the functional, $I = \int_0^{\frac{\pi}{2}} (y'^2 - y^2 + 4y \cos x) dx$, given that $y(0) = 0$, $y\left(\frac{\pi}{2}\right) = 0$.

$$y\left(\frac{\pi}{2}\right) = 0.$$

(07 Marks)

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18MAT753

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Advanced Mathematical Methods

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Reduce the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ to tridiagonal form and find its eigen values using

Given's method

(12 Marks)

- b. Find the eigen values of the following matrix by House holder method $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$

(08 Marks)

OR

- 2 a. Use the Birge Vieta method to find a real root correct upto three decimal places for the equation $x^3 - 11x^2 + 32x - 22 = 0$. Take $P_0 = 0.5$. (12 Marks)
- b. Using 'Bairstow' method, find the quadratic factor of $x^3 + x^2 - x + 2 = 0$ by taking $P = -0.9$, $q = 0.9$. (08 Marks)

Module-2

- 3 a. Solve $u_{xx} + u_{yy} = 0$ for the following square with the boundary conditions as indicated in the Fig Q3(a).

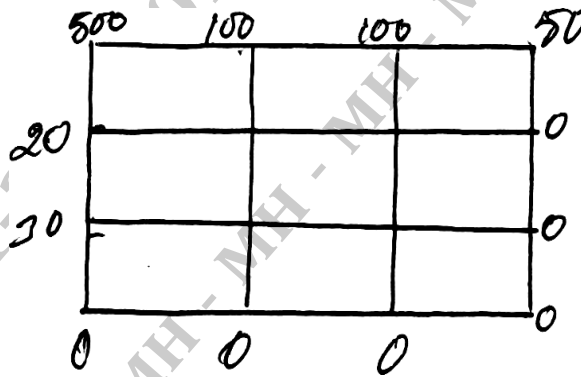


Fig Q3(a)

(10 Marks)

- b. Solve the Poisson equation $u_{xx} + u_{yy} = 8x^2y^2$ for the following square mesh with $u(x,y) = 0$ on the boundary and mesh length equal to 1.

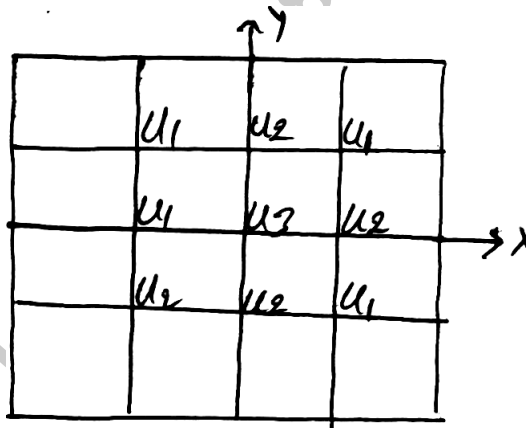


Fig Q3(b)

(10 Marks)

OR

- 4 a. Solve the wave equation $u_{tt} = 4u_{xx}$ subjected to $u(0, t) = 0$, $u(4t) = 0$, $u_t(x, 0) = 0$ and $u(x, 0) = x(4 - x)$ by taking $h = 1$, $k = 0.5$. (10 Marks)
- b. Using Crank – Nicholson method, solve $u_{xx} = 16 u_t$, $0 < x < 1$, $t > 0$, given $u(x, 0) = 0$, $u(0, t) = 50t$. Compute u for two steps in the direction of t by taking $h = 0.25$. (10 Marks)

Module-3

- 5 a. Find the Taylor's series expansion of $f(z) = \frac{1}{z^2 + 4}$ about $z = -i$. (07 Marks)

b. Evaluate $\int_0^{2\pi} \frac{d\theta}{5 - 3\cos\theta}$. (07 Marks)

c. Find the residue of $f(z) = \frac{ze^2}{(z-a)^3}$ at its pole. (06 Marks)

OR

- 6 a. Find the Laurent's series expansion of $f(z) = \frac{1}{(z+1)(z+3)}$ in $1 < |z| < 3$. (07 Marks)

b. Using Cauchy's residue theorem, evaluate $\int_C \frac{(4-3z)dz}{z(z-1)(z-2)}$, where C is the circle $|z| = \frac{3}{2}$. (07 Marks)

c. Evaluate $\int_0^\infty \frac{dx}{(1+x^2)^3}$, using contour integration. (06 Marks)

Module-4

- 7 a. A coin is tossed 900 times and head appears 490 times, can it be concluded that the coin is biased at 5% level of significance. Given $z_{0.05} = 1.96$. (07 Marks)
- b. An examination was conducted to two groups of students of two colleges. The following is the given data :
 College A : $\mu_A = 75, \sigma_A = 9, n_1 = 90$
 College B : $\mu_B = 79, \sigma_B = 7, n_2 = 60$
 Is there a significant difference in the performance of two colleges at 5% level of significance. Given $z_{0.05} = 1.96$. (07 Marks)
- c. The following table represent the production of three varieties of crops A, B, C, each grown on 4 plots. Find whether the variety differences are significant. Given $F_c = 4.26$.

Production/Acre			
Plot	A	B	C
1	6	5	5
2	7	5	4
3	3	3	-3
4	8	7	4

(06 Marks)

OR

- 8 a. A die is thrown 9000 times. The numbers 3 or 4 appears 3240 times. Test whether the die is unbiased at 5% level of significance. Given that $z_{0.05} = 1.96$. (07 Marks)
- b. Two independent samples of sizes 7 and 6 have the following data :
 Sample A : 28 30 32 33 31 29 34
 Sample B : 29 30 30 24 27 28 -
 Examine whether the samples have been drawn from a normal population having the same variance. Given $F_{0.05} = 4.39$. (07 Marks)
- c. Write a short note on :
 i) Null hypothesis ii) Types of errors iii) level of significance. (06 Marks)

Module-5

- 9 a. Show that the process $x(t) = A \cos (wt + \theta)$ is wide sense stationary. Where A and w are constants and θ is uniformly distributed random variable in $(0, 2\pi)$. (07 Marks)

- b. Find the fixed probability vector of the regular Stochastic matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$. (07 Marks)
- c. In a railway yard, trains arrive at a rate of 30 trains per day. Assuming that, the inter arrival time and service time distribution are exponential with an average of 36 minutes. Find :
 i) The probability that the yard is empty ii) average queue length, if the line capacity of the yard is 9 trains. (06 Marks)

OR

- 10 a. Three boy A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C, but C is just likely to throw the ball to B as to A. If C is the first to throw the ball, find the probabilities that for the fourth throw
i) A has the ball ii) B has the ball iii) C has the ball. (07 Marks)
- b. Cars arrive at a petrol pump, having one petrol unit, in a Poisson Fashion, with an average of 10 cars per hour. The service time is exponentially distributed with a mean of 3 minutes. Find : i) average number of cars in the system ii) average queue length iii) probability that the number of cars in the system is 2. (07 Marks)
- c. The auto correlation function for a stationary process is given by $R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$. Find the mean and variance of the process $X(t)$. (06 Marks)

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